

Auditing Generalization in AI-Generated Video Detection: A Six-Control Protocol and the VidAudit Toolkit

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Abstract

AI-generated video detection benchmarks such as GenVid-Bench and AIGVDBench are the de facto leaderboards, yet most evaluation protocols leave uncontrolled confounds that can inflate reported generalization. As an existence proof, a three-feature clip-length classifier reaches a leave-one-generator-out (LOGO) AUC of 0.998 on GenVidBench under unaudited evaluation, while measuring nothing about motion. A 20-paper survey finds none applying all six standard controls that would catch this, so we combine them into an audited protocol and apply it to six representative feature sources (three published detectors and three repurposed signal sources), re-running it cross-dataset on AIGVDBench. The audit both debunks and certifies: the trivial classifier collapses to near chance (0.529), a CLIP baseline is caught carrying dataset identity, and the 2025 forensic detector WaveRep clears the floor at out-of-distribution LOGO AUC 0.996 with chance-level real-vs-real coherence. At a deployable FPR of 0.1%, multiple high-AUC methods fall to single-digit recall and the leaderboard order changes, so we recommend an audited tuple (AUC, above-floor margin, operating-point recall, and calibration) over a single number. As a white-box positive control, we add TemporalSpec (codec motion vectors); via cross-substrate feature fusion (XSFF), a second substrate adds genuine complementarity that survives the audit. We release VidAudit, to our knowledge the largest unified and audited detector collection for this task, providing 14 detectors behind one plugin API, a leaderboard, and Croissant metadata, available at <https://github.com/KurbanIntelligenceLab/vidaudit>. Together, the protocol and toolkit move evaluation from leaderboard rank toward whether a result measures what it claims.

1. Introduction

For humans, discerning authentic video from AI-generated video has become so difficult, and so consequential in crit-

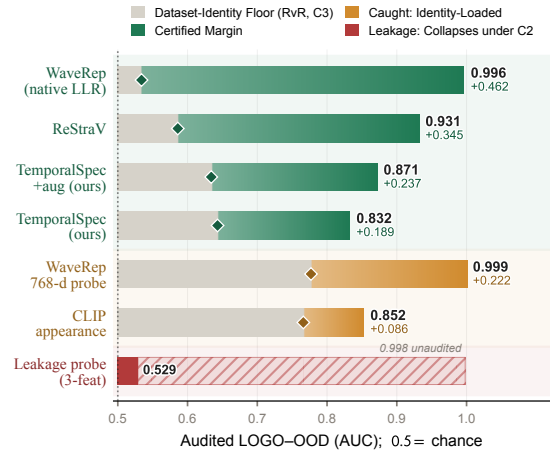


Figure 1. **The audit’s verdict at a glance.** Each bar is a detector’s audited LOGO OOD on the matched 27k GenVidBench subset, split into the dataset-identity floor it shares with a real-vs-real probe (gray; C3) and the above-floor margin (colored). Green detectors clear the floor by a wide margin and are *certified*; orange readouts sit close to high floors; the three-feature clip-length probe (red) collapses from 0.998 to chance under the $K=12$ filter (C2). Same numbers and verdict details as Table 2.

ical areas such as politics and finance [4, 30], that we require automated detectors to distinguish the two classes. The current challenge in this nascent field is that the existing and growing number of detectors is not comparable in any useful way, depending on disparate features whose *prima facie* scores, while commonly AUCs, are unlikely to be all equally good. Benchmark data sets exist; for example, GenVidBench [26] includes 6.78M videos and AIGVDBench [23] 440K. The body of generators is growing as well [3, 7, 15, 18, 25]. But a simple classifier, for example, utilizing a clip’s raw clip-length statistics reaches a leave-one-generator-out (LOGO) area under the ROC curve (AUC) of 0.998 on GenVidBench (Sec. 4.1). Under an unaudited evaluation, a simple clip-length meter is indistinguishable from a genuine detector. Additionally, the different features used in the collection of detectors do not con-

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trol for *confounds*: incidental properties of the data, *e.g.*, clip length, codec settings, or the corpus properties themselves. Because a classifier exploits these properties without identifying generation artifacts, the evaluation obscures how detection is working. While comparing detectors is necessary to judge relative performance, each is evaluated under a different head, data split, clip subset, dataset, and evaluation methodology, and thus scores, while appearing comparable, are *de facto* unnormalized.

We survey 20 recent (2024–2026) detection papers for six standard *controls*, each a check that removes or measures one confound so that a reported AUC reflects detection rather than an artifact of the data or harness: canonical codec re-encoding (C1), a leakage audit against a trivial baseline (C2), real-vs-real (RvR) coherence probing (C3), matched-harness re-training (C4), multi-seed and bootstrap confidence intervals (CIs; C5), and a true cross-dataset evaluation (C6).

With these controls we can now not only compare performance and gain insight into how the detectors are working, but also provide a path to building better detectors. We combine the six controls into an *audited evaluation protocol* applying it uniformly to six representative feature sources, with one source selected from each detector family. These include three published detectors, D3 [49], ReStraV [17], and the 2025 forensic detector WaveRep [9], as well as three signal sources repurposed as detectors, FVMD [21], RAFT [36], and CLIP-ViT-B/32. Under the audit on GenVidBench, the trivial clip-length classifier falls to chance (0.529), a CLIP appearance baseline is caught carrying dataset identity, WaveRep is certified against the audited confounds, and a motion-content detector holds at 0.832 (Fig. 1). Ranking the leaderboard by recall at a deployable operating point, a false-positive rate (FPR) of 0.1%, rather than by AUC, changes the order and drops several high-AUC methods to single-digit recall. We therefore report an audited tuple (AUC, margin above the real-vs-real floor, operating-point recall, and calibration) instead of a single metric.

Transparency improves the trustworthiness of an audit. Even a detector that clears every control, as WaveRep does, can only be trusted, not confirmed. The controls rule out the confounds we anticipated, but a black box could still be exploiting one we never tested, and we cannot look inside to check. We therefore introduce a *white-box positive control*, a codec-motion detector named TemporalSpec, whose 13 features are individually interpretable. Because each feature is interpretable, we can check its certification one feature at a time rather than take it on trust. We refer to the input domain that a detector reads as its *substrate*. TemporalSpec’s substrate is codec motion vectors, whereas CLIP-like detectors read deep appearance features. The clip-length leak described above is itself a codec-level artifact, which we iden-

Table 1. **No recent detector or benchmark applies all six controls** (representative rows; full 20-paper survey in the supplementary material). ● present, ◐ partial, ○ absent. C1 canonical re-encode, C2 leakage audit, C3 real-vs-real probe, C4 matched harness, C5 seeds/CIs, C6 cross-dataset.

	C1	C2	C3	C4	C5	C6
ReStraV [17]	○	○	○	○	○	●
FVMD [21]	○	○	◐	○	○	○
D3 [49]	○	○	○	◐	○	●
WaveRep [9]	●	◐	○	●	○	●
AIGVDBench [23]	●	◐	○	◐	○	◐
RobustSora [44]	○	●	○	●	●	◐
VidAudit	●	●	●	●	●	●

tified by inspecting codec metadata while building TemporalSpec. In addition, fusing the codec channel with an orthogonal substrate through cross-substrate feature fusion (XSFF) allows the audit to verify that a second substrate genuinely improves detection, rather than simply adding model capacity. We present TemporalSpec and XSFF as controls rather than as the paper’s primary detectors, since stronger published detectors achieve higher cross-generator AUC than TemporalSpec (Table 3).

Contributions. (i) We identify a previously unmeasured leakage channel. A classifier using only three clip-length features achieves a LOGO AUC of 0.998 on GenVidBench under unaudited evaluation, but drops to 0.529 under the audit. (ii) We introduce a six-control audited evaluation protocol (C1–C6), of which the 20 surveyed papers apply only a subset. (iii) We re-evaluate six representative feature sources, one per detector family, under a single matched protocol (three published detectors, D3, ReStraV, and WaveRep; three repurposed signal sources, FVMD, RAFT, and CLIP), placing methods that were never measured the same way on a common footing. (iv) We provide a white-box positive control, TemporalSpec, built from 13 interpretable codec-motion features, to show that the audit is constructive. With XSFF, we fuse this codec-motion channel with a DINOv2 deep-feature trajectory and show that the two substrates are genuinely complementary, a gain that survives the audit and replicates on AIGVDBench. (v) We release the evaluation as a reusable open-source toolkit built around *fourteen* AI-generated-video detectors, covering forensic, appearance, motion, codec, fusion, and multimodal-LLM families, behind a single plugin API. A new method can be scored under the full audit with one command. To our knowledge, this is the largest unified and audited detector collection for this task. The toolkit also includes the audited leaderboard, Croissant [1] metadata, and a re-scoring entry point (Sec. 4.5).

2. Related Work

Video generation and temporal evaluation. Diffusion- and transformer-based video generators [3, 7, 15, 18, 28]

differ substantially in their training corpora (e.g., HD-VG130M [42]), resolution, and frame rate, with the resulting generated-video diversity documented at scale by prompt-gallery datasets such as VidProM [41]. As a result, they produce heterogeneous spatiotemporal signatures. Classical quality metrics largely inherit their design from image generation and focus on frame-level fidelity. Fréchet Video Distance (FVD) and Fréchet Inception Distance (FID) [14, 37] can miss subtle temporal discontinuities [12], and FVD’s Fréchet distance over I3D (Inflated 3D ConvNet) features is computationally expensive and remains insensitive to fine-grained motion artifacts.

Motion vectors in machine learning. Modern video codecs such as H.264 and HEVC [34] use inter-frame motion compensation across I, P, and B frames, leaving motion vectors (MVs) as an abundant bitstream byproduct that can be read at negligible cost [31, 33]. Wu et al. [47] introduced deep compressed-domain action recognition, feeding codec motion vectors and residuals directly into CNNs without optical flow; Shou et al. [32] refined these coarse motion vectors into more discriminative, flow-like cues. Later work used MV features for video-quality enhancement, temporal grounding, and compression-aware super-resolution [10, 43, 50]. We adopt the same compressed-domain perspective, but apply it to real-vs-AI detection using a small set of hand-engineered statistics and a linear readout. Because codec MV statistics reflect encoder generation and rate-distortion (RD) heuristics beyond motion, we re-encode every input through one canonical H.264 pipeline before extracting features.

AI-generated video detection. Detectors for AI-generated video can be grouped into four families. *Frame-feature* detectors, such as D3 [49] and DeCoF [22], build on CLIP or video-language backbones and model temporal structure over per-frame features, requiring a GPU forward pass per clip. *Cross-modal* detectors, such as ATSS [39] and CMTA [38], compare visual and text-trajectory similarity from vision-language models; ATSS flags the unnaturally high cross-modal self-similarity of prompt-driven generation, while CMTA targets cross-modal temporal artifacts. Recent audio-visual and motion-depth fusion methods follow a similar principle to our XSFF, although they use different substrate pairs. *Reasoning* detectors, such as VidGuard-R1 [27], fine-tune a multimodal large language model (MLLM) with reinforcement-learning rewards and achieve top zero-shot accuracy, but require an MLLM-scale forward pass for each clip. *Motion-only* detectors remove appearance information entirely and test whether motion alone is discriminative. FVMD [21] histograms the velocity and acceleration of pre-trained point-tracker keypoints, a construction inspired by histogram-of-oriented-gradients (HOG) descriptors, whereas our work uses codec motion vectors. Recent frequency-domain and latent-

dynamics methods, including FreqNet-style spectral filtering and StyleGRU-style temporal latents, operate on substrates that we do not benchmark here but are candidate channels for future XSFF extensions. Two recent detectors, D3 and ReStraV [17], share our intuition: the *shape* of temporal change carries a generative fingerprint. Both methods characterize the curvature or second-order statistics of a deep-encoder trajectory, using XCLIP or DINOv2. Our temporal-structure features (Tier 1; Sec. 3.3) measure closely related quantities, including acceleration kurtosis, spectral slope, and autocorrelation (ACF) decay, but operate on the raw codec macroblock motion field already present in the bitstream rather than on a deep-feature trajectory. These detector families occupy different cost-accuracy regimes. RAFT-Large requires ~ 400 ms per clip on an A40, XCLIP-ViT-B/16 has a comparable cost, and codec-MV inference runs on a CPU in tens of milliseconds. We therefore position our compressed-domain detector not as a competitor to D3 or ReStraV in raw accuracy, but as an inexpensive complementary channel that can run alongside them. We report direct numerical comparisons with FVMD, RAFT, D3, and ReStraV on a matched 27,000-clip subset, and evaluate cross-dataset generalization on AIGVDBench [23], an independently curated benchmark.

3. Method

The audited evaluation protocol applies six controls to each detector we compare (Sec. 3.1). To exercise it, we build TemporalSpec, a white-box detector that reads codec motion vectors (Sec. 3.2) and summarizes them as a compact family of temporal-spectral features (Sec. 3.3). XSFF extends TemporalSpec by fusing these features with a DINOv2 deep-feature trajectory (Sec. 3.4). We evaluate on two fixed subsets of GenVidBench. The *matched* subset (27,000 clips, balanced at 3k per source across 7 generators plus Vript and HD-VG-130M) is the comparison subset for all detectors, so no source’s over-representation skews it; the *full* subset (116,066 clips, Pair1 $\cup$ Pair2) serves only the TemporalSpec multi-seed check.

3.1. Audited evaluation protocol

The protocol consolidates six controls applied uniformly to every comparison row. (C1) *Canonical re-encode*: every input passes through one H.264 pipeline before feature extraction, removing codec-side fingerprints. (C2) *Leakage-audited K -non-I-frame filter*: a trivial-baseline classifier (here, three frame-count features) is fit on the uncontrolled pipeline to measure an upper bound on leakage. The filter then keeps a fixed number of non-I-frames ($K=12$) per clip. This equalizes clip length and removes the frame-count confound the baseline exploits. (C3) *Real-vs-real coherence probe*: a classifier is trained to discriminate the two real sources, giving a measurable floor on dataset-

bias contribution. (C4) *Matched-harness comparison*: every compared detector is re-fit under the same readout (an L2-regularized logistic regression, L2-LR), the same 80/20 split, the same LOGO protocol, and the same 27,000-clip subset. (C5) *Multi-seed stability and bootstrap CIs*: every principal number is reported with its multi-seed stability and a bootstrap confidence interval (5 seeds, $B=1000$ percentile bootstrap). (C6) *Cross-dataset validation*: the audit is re-run on AIGVDBench, a benchmark whose generator roster, real-clip pool, and recording pipeline were curated separately from GenVidBench. Our survey of 20 recent (2024–2026) AI-generated-video detection papers (representative rows in Table 1; full table in the supplementary material) finds no prior work combining all six. The canonical re-encode (C1) appears in full in only 2 of 20 papers, the leakage audit (C2) and multi-seed/bootstrap CIs (C5) in only one each, and the real-vs-real coherence probe (C3) in none. Sec. 4.1 measures the consequence of skipping them. We refer to a detection result as *audited* when all six controls have been applied.

3.2. Codec motion vectors

We use the motion vectors that standard video codecs (H.264/AVC, HEVC) produce during inter-frame prediction. They are emitted during decoding at negligible additional cost and, unlike optical flow, need no separate motion-estimation procedure. A video is represented as a sequence of T frames, $\mathcal{V}=\{F_1, \dots, F_T\}$, where $F_t \in \mathbb{R}^{H \times W}$. Each frame is partitioned into non-overlapping $M \times M$ macroblocks, typically of size 16×16 , indexed by (i, j) , with N_{MB} macroblocks per frame. Each frame also has one of three coding roles. Intra-coded I-frames use no temporal reference, predictive P-frames reference past frames F_{t-k} , and bidirectional B-frames reference both past and future frames $F_{t \pm k}$. P- and B-frames carry motion vectors and residuals, whereas I-frames carry neither. For each inter-coded macroblock, the encoder searches over a discrete window of pixel offsets to find the best matching reference block [31]. This process yields a motion vector $\mathbf{mv}_t^{(i,j)} \in \mathbb{Z}^2$, measured in pixels. We define the corresponding motion magnitude as $m_t^{(i,j)} = \|\mathbf{mv}_t^{(i,j)}\|_2$. Since I-frames do not contain motion vectors, we treat them as frames with zero motion.

Codec MVs as a structured operator on motion. We model the MV field as the output of a deterministic, RD-constrained operator applied to the true pixel-level displacement $\mathbf{u}_t$. We define $\Phi = \mathcal{S}_\lambda \circ \mathcal{Q}_\delta \circ \mathcal{L}_M$ [34, 46], where $\mathcal{L}_M$ performs block averaging, $\mathcal{Q}_\delta$ applies integer-pixel quantization with a dead-zone, and $\mathcal{S}_\lambda$ represents RD-based selection that favors smaller-magnitude vectors; the action of each component is formalized in the supplementary material. We assume, as a *modeling choice* rather than a de-

rived identity, that Φ acts as a perceptual filter that retains coarse, large-magnitude motion structure while suppressing sub-block jitter, sub-pixel flicker, and pixel-domain noise.

Complexity. Feature computation is a single pass over the macroblock grid, with complexity $\mathcal{O}(T \cdot N_{\text{MB}})$. On a single CPU core it takes a mean of 14.2 ms per clip and a median of 21.7 ms, and the runtime is bimodal, near 1 ms for short clips and 22 ms for longer ones. The canonical H.264 re-encode is a separate one-time preprocessing pass, with a single-core median of ~ 2.5 s per clip on foreign-codec inputs (full per-stage timings in the supplementary material); clips that already arrive in canonical H.264 skip it. When a corpus is normalized once and screened offline, the per-clip cost is therefore the ~ 14 ms mean. An online deployment without pre-canonicalization carries the re-encode as an upstream cost, as does any detector that must remove codec-side fingerprints.

3.3. Temporal feature family

We organize features into two tiers. Tier 0 captures *how much* motion is present; Tier 1 captures *how* that motion is structured over time.

Why real and generated MV spectra differ. Let $r(t)$ be the mean MV-magnitude time series of a spatial region (a lattice cell, defined below) and $S_r(f)$ its power spectral density (PSD), where f is temporal frequency in cycles per frame, ranging from 0 to $1/2$ (the Nyquist limit). Generated video should carry less high-temporal-frequency energy than real footage. Two mechanisms predict this difference: the neural spectral bias toward low frequencies [11, 29, 40], and the low-pass action of short 3D-convolution and temporal-attention kernels (cutoff $f_c \ll 1/2$). The supplementary material develops both arguments. For a low reference frequency $f_0 < f_c < f$, both mechanisms predict

$$\frac{S_r^{\text{gen}}(f)}{S_r^{\text{gen}}(f_0)} < \frac{S_r^{\text{real}}(f)}{S_r^{\text{real}}(f_0)}. \quad (1)$$

The codec operator Φ amplifies this spectral contrast by suppressing the pixel-domain noise floor before the features are computed. Eq. (1) predicts the direction in which three Tier 1 features should differ for generated video, namely a steeper spectral slope, lower spectral flatness, and a longer ACF decay time, together with one empirically observed correlate, lower acceleration kurtosis. The first three follow from Eq. (1) only heuristically, and the kurtosis direction is empirical rather than a theoretical consequence. The supplementary material states the four inequalities, gives the full derivation, and argues that our features are orthogonal to the deep-feature trajectories of D3 [49] and ReStrav [17].

Feature definitions. Let $\bar{m}_t = \frac{1}{N_{MB}} \sum_{i,j} m_t^{(i,j)}$ denote the mean MV magnitude in frame t . Tier 0 computes four clip-level scalars: **(T0-1)** the mean of $\bar{m}_t$, **(T0-2)** the sparsity ratio, **(T0-3)** the variance $\text{Var}(\bar{m}_t)$, and **(T0-4)** the mean temporal difference $|\bar{m}_t - \bar{m}_{t-1}|$. Tier 1 captures localized temporal structure by partitioning the macroblock grid into a 4×4 lattice of 16 cells, justified by the supplementary lattice sweep. For each cell, $r(t)$ is the per-frame mean of $m_t^{(i,j)}$, and its PSD $S_r(f)$ is estimated with Welch’s periodogram [45], or, for short clips, a fourth-order autoregressive (AR) model fit by Burg’s method, a parametric estimator more reliable than the periodogram on short series. From $S_r(f)$ and $r(t)$ we compute six per-cell statistics: **(T1-1)** the spectral slope from a log–log fit of $S_r(f)$, **(T1-2)** the spectral flatness, **(T1-3)** the ACF decay time τ_0 , **(T1-4)** the first ACF zero-crossing lag, **(T1-5)** the excess kurtosis of the acceleration $a_t = r(t) - 2r(t-1) + r(t-2)$, and **(T1-6)** its skewness. Across the 16 cells, each statistic contributes its median, giving six features, and T1-1 to T1-3 additionally contribute their interquartile range (IQR), giving three more, for nine Tier 1 features. With the four Tier 0 scalars, this yields a 13-dimensional feature vector. Exact definitions and the median/IQR selection are given in the supplementary material, which also verifies seed stability, with the multi-seed detection-AUC standard deviation ≤ 0.005 at $K=12$ (the best trade-off in the K/AR -order/max-lag sweep).

Frame-count uniformity. The PSD estimator, ACF max-lag, and per-feature IQRs all depend on the clip length T . Because GenVidBench generators use different clip lengths, this length variation can introduce a spurious frame-count fingerprint in the features. We therefore enforce exactly $K=12$ non-I-frames per clip after motion filtering, which places all clips on the same Burg-AR(4) PSD configuration. The leakage audit in Sec. 4.1 verifies that this removes the dominant frame-count confound. Sensitivity to K , AR order, and ACF max-lag is reported in the supplementary material.

Codec pipeline equivalence. Because GenVidBench clips use heterogeneous native codecs and frame rates, and codec MVs depend on encoder configuration [34], we equalize the encoding conditions before MV extraction. Each clip is re-encoded with a canonical H.264 pipeline and rescaled to a fixed spatial resolution. The macroblock grid is deterministic, so any content-dependent residual introduced by RD search should appear in the real-vs-real coherence probe (C3). This probe yields a real-vs-real coherence AUC of 0.628 on the 116k subset and 0.643 on the 27k matched subset (Sec. 4.1), the dataset-identity floor that any detector must clear. Full pipeline parameters and robustness to post-processing and codec variation are provided in the supplementary material.

3.4. Two-channel detection with cross-substrate feature fusion (XSFF)

XSFF fuses two motion-evidence channels. The first is the codec-MV branch; the second is the published ReStraV detector [17], which encodes per-frame DINOv2 ViT-S/14 embeddings into a 21-d trajectory vector. We denote the codec-motion feature map by $\varphi_{MV}(v) \in \mathbb{R}^{13}$ and the deep-trajectory feature map by $\psi_{DINO}(v) \in \mathbb{R}^{21}$. These are distinct from the codec operator Φ in Sec. 3.2. Assuming conditional substrate independence given the generator class, the sum of per-substrate log-likelihood ratios (LLRs) is the Bayes-optimal estimator of the joint LLR. We audit this assumption empirically in the supplementary material, where the marginal dependence is small: $\hat{I}(\varphi_{MV}; \psi_{DINO}) = 0.057$ nats. XSFF implements this fusion with an L2-regularized logistic readout on the concatenated 34-dimensional feature vector: $\text{XSFF}(v) = \sigma(\mathbf{w}^\top [\varphi_{MV}(v) \oplus \psi_{DINO}(v)] + c)$, where σ is the logistic function and $\mathbf{w}, c$ are fit on each LOGO training fold. Because any fixed linear blend of the two branch scores is itself a linear function of the concatenated features, the feature-level readout can represent every such blend as a special case (proof in the supplementary material), so its fit on the training data matches or beats the best blend. This guarantee is in-sample only. The added flexibility can overfit, so a lower training risk need not yield a cross-generator (out-of-distribution) gain. Empirically, however, the joint readout transfers to held-out generators. On LOGO OOD, XSFF exceeds a fixed- α blend, with α tuned without test access, by +1.06 AUC points ($p=0.045$, 6/7 generators, $n=7$).

4. Results: audited re-evaluation

Results are reported under the audited protocol in Sec. 3.1: canonical H.264 re-encoding (C1), the $K=12$ non-I-frame filter (C2), matched-harness L2-LR (C4), and multi-seed plus bootstrap CIs (C5). On GenVidBench [26], Text2Video-Zero is structurally excluded because too few non-I-frames survive. On AIGVDBench, AccVideo and HunyuanVideo are retained under a relaxed $K=8$ filter (Sec. 4.4). LOGO ID denotes AUC on held-out clips from the training generators, and LOGO OOD denotes AUC on held-out generators.

4.1. What the audit catches and certifies

Table 2 separates the audit findings into two cases. *First*, a trivial three-feature L2-LR using raw I-, P-, and B-frame counts reaches LOGO AUC 0.998 on GenVidBench under unaudited evaluation, and a single-feature LR on n_{total} alone reaches ~ 1.0 . These values resemble state-of-the-art performance, but they are driven by clip length. This probe shows that the confound exists in the data, not that it explains any specific leaderboard result. GenVidBench’s released protocol samples a fixed number of frames, normal-

Table 2. **What the audit catches and certifies.** Audited LOGO OOD on the matched 27k GenVidBench subset. *Top*: clip-length leakage probes collapse to chance under the $K=12$ filter (C2; unaudited score in parentheses). *Middle*: CLIP and a raw 768-d probe of WaveRep’s backbone pass C2 but sit just above high real-vs-real floors (C3). *Bottom*: motion-content detectors and WaveRep’s forensic readout clear the floor by a wide margin and are *certified*. Full per-detector RvR and cost in Table 3.

	dim	Audited LOGO OOD	RvR floor / margin
<i>Leakage probes (collapse under C2):</i>			
Single-feature (n_{total} , clip length)	1	~chance (was ~1.000)	no floor: caught by C2
Three-feature (n_I, n_P, n_B , frame counts)	3	0.529 (was 0.998)	
<i>Caught carrying dataset identity (pass C2, high RvR):</i>			
CLIP-ViT-B/32 mean-pool (appearance)	13	0.852	0.766 / +0.086
WaveRep DINOv2 raw-feature probe	768	0.999	0.777 / +0.222
<i>Certified (well above the floor):</i>			
TemporalSpec (13-d codec-motion, ours)	13	0.832	0.643 / + 0.189
TemporalSpec+aug (20-d, LightGBM, ours)	20	0.871	0.634 / + 0.237
ReStraV DINOv2 ViT-S/14 (matched harness)	21	0.931	0.586 / + 0.345
WaveRep native LLR [9]	3	0.996	0.534 / + 0.462

izing clip length before its own models process the input. However, no surveyed protocol explicitly audits this confound, so pipelines that ingest variable-length clips remain exposed. Under the audit’s $K=12$ filter (C2), both probes collapse to near chance, while the motion-content detector TemporalSpec holds at 0.832.

Second, the real-vs-real coherence probe (C3) lower-bounds the dataset-identity contribution at 0.643 on the matched subset. CLIP-ViT-B/32 appearance reaches a strong LOGO OOD AUC of 0.852, but its RvR is 0.766, leaving a margin of only +0.086. Thus, most of its cross-generator signal comes from real-source identity. In contrast, TemporalSpec gives 0.832 versus RvR 0.643, for a margin of +0.189, while ReStraV [17] gives +0.345. The audit therefore catches two distinct confounds, clip-length leakage (C2) and the dataset-identity floor (C3), either of which can be hidden by a single unaudited AUC.

The certified white-box control. TemporalSpec occupies the certified end of Table 2. Its 13-feature L2-LR reaches audited LOGO OOD AUC 0.832 at RvR 0.643 on the 27k subset, and 0.819/0.628 on the 116k full subset. Performance is stable across seeds, with standard deviation ≤ 0.005 (supplementary material). The signal is interpretable and does not reduce to “real simply moves more”: mean MV magnitude alone is near chance on five of seven generators, the 10-feature magnitude-free variant still spans [0.81, 0.93] per generator, and Tier 1 spectral features add up to 11 AUC points over Tier 0 magnitude.

LOGO and real-vs-real coherence (C3). Bootstrap 95% CIs ($B=1000$; full per-detector table in the supplementary material) place the RvR floor (C3) at 0.643 [0.611, 0.675] and TemporalSpec ID detection at 0.854 [0.843, 0.866]. Since every per-generator LOGO AUC exceeds the dataset-identity floor, detection does not reduce to a real-source fingerprint. The Tier 0/Tier 1 split is further supported

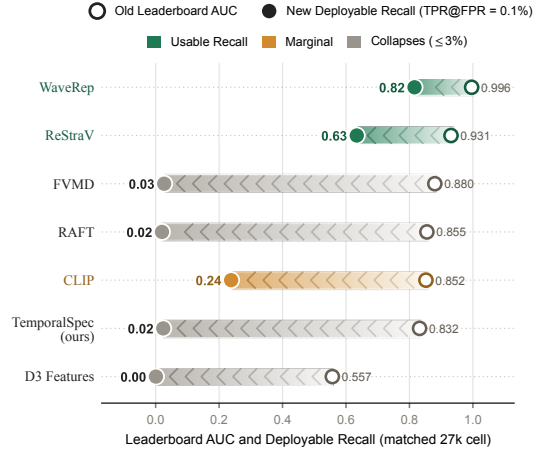


Figure 2. **The leaderboard metric does not predict deployability.** For each detector, the hollow marker is its old leaderboard LOGO OOD AUC and the bold filled marker its new deployable recall (true-positive rate, TPR, at FPR= 0.1%) on the matched 27k subset; the connector traces the drop. WaveRep and ReStraV (green) remain effective at this operating point; the high-AUC published baselines (gray) collapse to $\leq 3\%$ recall, and the order changes, with CLIP retaining an order of magnitude more recall than FVMD and RAFT at a *lower* AUC. Same numbers as Table 4.

by a 1,071-pair within-prompt concordance test against human VBench [16] judgments. Tier 1 tracks human motion-smoothness ratings, while Tier 0 shows the opposite pattern. Full results and the scale-invariance derivation are provided in the supplementary material.

4.2. Audited re-evaluation of six recent detectors

Table 3 re-evaluates six baselines under the audited matched-harness L2-LR protocol. The cost-vs-OOD frontier is reported in the supplementary material. We re-fit each detector with a common readout to compare *representations* on equal footing, and report the native head beside the re-fit row. A gap between the two indicates that the detector’s power lies in its head rather than in its representation.

FVMD [21]: the full 1024-d representation reaches LOGO OOD 0.880 with the lowest RvR in the table (0.574). Reducing it to 13 dimensions lowers OOD to 0.863 with PCA and 0.761 with top-13 features. The audit confirms genuine motion-content discrimination. **RAFT [36]:** replacing codec MVs with RAFT-Large flow in the same 13-d pipeline gives 0.855, compared with 0.832 for codec MVs, at unchanged ID. Thus, the Tier 0/1 features, rather than the motion source, carry the signal. The readout sweep remains in 0.820–0.849 (supplementary material). **D3 [49]:** the published training-free head reaches 0.887 OOD, but its 768-d XCLIP-ViT-B/16 representation falls to 0.557 under our L2-LR readout, a loss of 33 AUC points. D3’s power therefore resides in the *head*, not in the representation. **Re-**

Table 3. **Re-evaluation under the audited protocol on the matched 27,000-clip GenVidBench subset.** One representative detector per family (FVMD, RAFT, D3, ReStraV, CLIP, WaveRep); pure ablations and all fusion variants are in the supplementary material. Every row uses the same L2-LR matched harness (C1–C5) at a fixed seed, 80/20 split, LOGO mean over 7 held-out generators. Lower RvR is better (chance 0.5; the dataset-identity floor); the *margin* column (= LOGO OOD – RvR) is the genuine signal above that floor. Cost (ms/clip) is per-clip post-decode inference (* = order-of-magnitude estimate; “14+10*” = CPU codec pass + GPU ReStraV forward pass; canonical re-encode excluded).

Method	Subset	dim	ms/clip	LOGO ID	LOGO OOD	RvR	margin
<i>Our case-study detectors:</i>							
TemporalSpec (ours, 13-d, L2-LR)	27k	13	14 (CPU)	0.909	0.832	0.643	+0.189
TemporalSpec+aug (ours, 20-d, LightGBM)	27k	20	15 (CPU)	0.960	0.871	0.634	+0.237
<i>One published detector per family (Sec. 2); variant ablations in the supplementary material:</i>							
FVMD (motion-distance, full 1024-d) [21]	27k	1024	59*	0.924	0.880	0.574	+0.306
FVMD (dimensionality-matched, PCA-13) [21]	27k	13	59*	0.880	0.863	0.555	+0.308
RAFT (optical-flow → Tier 0/1) [36]	27k	13	400	0.901	0.855	0.627	+0.228
D3 (frame-pixel, features → L2-LR) [49]	27k	768	50*	0.668	0.557	0.546	+0.011
D3 (frame-pixel, native head) [49]	27k	1	50*	0.887	0.887	0.421	+0.466
ReStraV (perceptual-straightening) [17]	27k	21	10*	0.995	0.931	0.586	+0.345
CLIP-ViT-B/32 (appearance)	27k	13	67*	0.945	0.852	0.766	+0.086
WaveRep (forensic LLR, DINOv2 ViT-B/14) [9]	27k	3	420	0.996	0.996	0.534	+0.462
WaveRep (same backbone, raw-feature probe) [9]	27k	768	420	1.000	0.999	0.777	+0.222
<i>Cross-substrate fusion (all seven fusion variants in the supplementary material):</i>							
XSFF: MV ⊕ ReStraV (feature-level 34-d)	27k	34	14+10*	0.996	0.946	0.604	+0.342
fixed-α blend (best linear, baseline)	27k	n/a	14+10*	0.995	0.935	0.604	+0.331

StraV [17]: the 21-d straightening features reach 0.931 OOD at RvR 0.586 (published MLP head: 0.901; supplementary material), confirming genuine motion-content signal. **WaveRep [9]** fine-tunes a DINOv2 ViT-B/14 backbone (768-d) with wavelet-band forensic augmentation. It is the strongest method we re-evaluate, and its two readouts show both verdicts the audit can return on a single backbone. Its native per-frame LLR aggregate reaches LOGO OOD 0.996 at chance-level RvR (0.534), clearing the dataset-identity floor by +0.462; the audit therefore *certifies* it. In contrast, when the same DINOv2 ViT-B/14 backbone is used as a raw 768-d feature probe, it reaches a nominally higher 0.999 but raises RvR to 0.777. These raw appearance features carry Vript-vs-HD-VG-130M identity, which WaveRep’s forensic head discards. C3 therefore localizes where the leak enters. We evaluate WaveRep as a pre-trained detector, using its native calibrated score as in the D3/ReStraV-native rows, so the LOGO results measure pre-trained transfer. Because WaveRep is trained on a single generator, at most one of the seven held-out generators could be in-distribution. This is the only caveat on the zero-shot interpretation. **CLIP-ViT-B/32:** mean-pooled features, PCA-reduced to 13 dimensions to match TemporalSpec, reach 0.852 OOD and therefore match TemporalSpec on LOGO AUC. However, CLIP sits only +0.086 above its 0.766 RvR floor, making it the canonical example of why C3 is needed in addition to C2.

Where TemporalSpec/XSFF sit on the frontier. WaveRep, ReStraV, full FVMD, and D3’s native head all exceed TemporalSpec in cross-generator AUC. TemporalSpec instead provides the lowest-cost CPU-only point that clears

the RvR floor, by +0.189, at 14 ms/clip. The 20-d augmentation raises LOGO OOD to 0.871 with LightGBM. Fusion with ReStraV’s DINOv2 trajectory raises OOD to 0.946 at RvR near 0.60. This gain would disappear if codec motion were already subsumed by the appearance trajectory, so we interpret it as genuine complementarity rather than added capacity. Full fusion variants and significance tests are reported in the supplementary material.

4.3. Beyond AUC: what the leaderboard hides

AUC averages over all thresholds, including high-FPR regions irrelevant to deployed screening systems. Under the same audited LOGO subset, we therefore report operating-point TPR at FPR = 1% and 0.1%, partial AUC for FPR ≤ 10%, and calibration metrics, namely Brier score and expected calibration error (ECE). Fig. 2 and Table 4 show the AUC-vs-recall contrast and the full per-detector metric tuple. The AUC column hides three findings. **(i) Operating-point collapse:** at FPR=0.1%, only WaveRep (TPR 0.82) and ReStraV (TPR 0.63) remain usable, while FVMD, RAFT, and TemporalSpec fall to TPR ≤ 0.03 despite AUCs of 0.83–0.89. Thus, an AUC of 0.88 can correspond to only 3% recall at a deployable threshold. **(ii) Calibration does not track AUC:** FVMD (ECE 0.087) is better calibrated than the higher-AUC ReStraV (ECE 0.120), so AUC alone does not indicate whether a score can be interpreted as a probability. **(iii) The AUC order changes at the operating point:** CLIP is statistically tied with FVMD (paired bootstrap $p=0.34$) and RAFT ($p=0.36$) in AUC over the 7 generators, but at FPR=0.1% its recall (0.24) is roughly an order of magnitude higher than theirs (0.03 and 0.02). Thus, the leaderboard order changes under operating-point

Table 4. **The audited metric tuple, per re-evaluated detector.** LOGO-OOD on the matched 27k subset under a uniform L2-LR readout. pAUC@10%: partial AUC over $\text{FPR} \leq 0.10$; TPR@FPR: operating-point recall; Brier/ECE: calibration (lower better). Detectors are listed in descending order of audited AUC (Table 3); the ordering of the middle rows does *not* hold under the operating-point or calibration columns.

Detector	pAUC@10%	TPR@1%	TPR@0.1%	Brier	ECE
WaveRep native LLR	0.980	0.935	0.816	0.026	0.039
ReStraV	0.871	0.726	0.634	0.097	0.120
FVMD full	0.705	0.107	0.027	0.127	0.087
RAFT T0+T1	0.650	0.110	0.020	0.158	0.104
CLIP appearance	0.782	0.417	0.238	0.128	0.094
TemporalSpec (MV 13-d)	0.665	0.130	0.024	0.166	0.106
D3 features	0.479	0.001	0.001	0.277	0.177

metrics. Only the WaveRep/ReStraV top and the D3 bottom are stable across all columns. We therefore recommend reporting the audited metric *tuple*, rather than a single AUC, as part of the protocol.

4.4. Cross-dataset replication on AIGVDBench (C6)

We run the full audit on AIGVDBench [23], a 2025-era benchmark with a generator roster and real-source pool curated independently of GenVidBench. The roster includes five generators from the Open-Source split at $K=12$, plus AccVideo and HunyuanVideo under the $K=8$ relaxation (Table 5). All controls except C3, which requires multiple real sources, carry over. Classifiers are re-fit for each LOGO fold, and no GenVidBench weights are reused. On the harder 2025 roster, both single channels drop sharply. TemporalSpec’s 13-d L2-LR falls to LOGO OOD 0.602 at $K=12$ and 0.687 with augmentation. ReStraV falls to 0.678, and per-generator OOD drops to 0.195 on Open-Sora, well below chance. Thus, the audit exposes rather than hides the difficulty of detecting 2025 generators. Two observations still transfer. First, the cost-vs-OOO *ordering* from GenVidBench is preserved, with the CPU motion-vector channel below ReStraV in OOD AUC. Second, cross-substrate complementarity reappears directionally. The $\text{MV} \oplus \text{ReStraV}$ blend improves over ReStraV alone by +1.6 AUC points, compared with +1.4 on GenVidBench. We report this as corroboration, not as a significance claim at $n=7$, and make no deployable-accuracy claim for the 2025 generators.

4.5. Reproducibility artifacts

The toolkit packages the audit as a reusable benchmark: the six controls and matched-harness pipeline (C1–C6), the fourteen-detector collection, an audited leaderboard with subset-provenance labels, a standardized data package (canonical re-encoding, the K -non-I-frame filter, Croissant [1] metadata), and a re-scoring entry point for detector predictions.

Table 5. **Cross-dataset replication (C6): GenVidBench 27k vs. AIGVDBench.** Same audited L2-LR matched-harness pipeline; classifiers re-fit per benchmark (no GenVidBench weights reused). The cross-substrate *complementarity lift* (MV+ReStraV blend vs. ReStraV-alone) replicates directionally (+1.4 vs. +1.6 AUC points).

Detector (LOGO OOD)	GenVidBench 27k	AIGVDBench	Δ
TemporalSpec (13-d, L2-LR)	0.832	0.602 ($K=12$)	−0.230
TemporalSpec+aug (20-d, LightGBM)	0.871	0.687 (13-d, $K=8$, 7 gens)	−0.184
ReStraV (21-d, matched L2-LR)	0.930	0.678	−0.253
MV $\oplus$ ReStraV score blend	0.944 ($\alpha=0.25$)	0.694 ($\alpha=0.50$)	−0.250
<i>Complementarity lift (MV$\oplus$ReStraV blend − ReStraV-alone):</i>			
	+1.4 AUC points	+1.6 AUC points	same sign ✓

5. Limitations

The audit covers two benchmarks, GenVidBench and AIGVDBench. Detector families without released code or weights at evaluation time, including DeCoF [22], ATSS [39], CMTA [38], VidGuard-R1 [27], StyleGRU, and DeMamba [8], are surveyed in the supplementary material but not re-scored under the protocol. Detection is also uneven across generators; a single unseen generator (MuseV) sharply lowers TemporalSpec’s LOGO OOD, so an open-ended roster needs at least one appearance-domain channel. On AIGVDBench the real-vs-real probe (C3) cannot run, with only one real source, so the dataset-identity floor is inferred and the 2025 roster reported as directional replication, not deployable accuracy. The codec-operator model, spectral-bias derivation, and substrate-independence argument are modeling assumptions, as detailed in the supplementary material. They constrain the interpretation of the white-box control, not the audit itself, which is empirical. Like any fixed protocol, ours can be circumvented once its controls are known; the natural safeguard is an orthogonal-substrate channel [17]. *Broader impact:* detection is dual-use, so we release evaluation controls rather than an evasion tool, and for operational deployment we recommend gated release and monitoring.

6. Conclusion

AI-generated-video detection benchmarks often do not measure what they report. Under unaudited evaluation, a three-feature clip-length classifier reaches LOGO AUC 0.998 on GenVidBench, and none of the 20 surveyed papers applies controls that catch it. The audited protocol both debunks and certifies. It collapses that classifier to chance, catches two appearance probes carrying dataset identity, and certifies WaveRep well above the floor. TemporalSpec and XSFF show the audit is also constructive, with cross-substrate complementarity surviving the controls. Reporting an audited tuple (AUC, margin, recall, calibration) rather than a single AUC shifts attention from leaderboard rank to whether a result measures what it claims.

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Supplementary Material

This supplementary material is grouped as follows. *Validating the headline numbers*: the audited metric suite of operating-point TPR, partial AUC, and calibration (Sec. S4); the bootstrap 95% confidence intervals for every reported AUC (Sec. S5); and the multi-seed stability check (Sec. S21). *Per-generator and cross-dataset breakdowns*: the distribution-level FVMD per-generator ledger paired against our per-generator MV detection AUC (Sec. S6); the MuseV per-feature Cohen’s d analysis and leave-one-feature-out ablation (Sec. S10); the cross-pair generalization result (Sec. S23); and the within-AIGVDBench LOGO second-dataset check, which also replicates the MV + ReStraV complementarity finding (Sec. S26). *Robustness and sensitivity*: the full $K/p_{AR}/L$ sensitivity sweep (Sec. S7); the GOP and frame-type fingerprint probe that quantifies the clip-length leakage upper bound (Sec. S8); the full per-condition robustness sweep with the cross-codec measurability discussion (Sec. S9); and the lattice-resolution sweep (Sec. S19). *Why it works*: the Tier 0/Tier 1 scale-invariance derivation (Sec. S15) and the spectral-mechanism hypothesis (Sec. S16). *Comparisons and extensions*: the readout sweep across linear, tree-ensemble, kernel, and MLP heads on the same 13-feature set (Sec. S11); the D3 head-to-head on the matched subset (Sec. S12); the matched-harness reproduction detail for FVMD, RAFT, and D3 (Sec. S14); the MV + D3 score-ensemble sweep (Sec. S22); the augmented vector-field feature sweep that lifts LOGO OOD to 0.871 (Sec. S24); and the ReStraV head-to-head that reaches LOGO OOD 0.931 at the small-ViT GPU-forward operating point (Sec. S25). *Cost and deployment*: the end-to-end per-stage timing (Sec. S20); the RAFT cost profile (Sec. S27); and the deployment guidance plus reproducibility manifest (Sec. S13). Section, equation, table, figure, and reference numbering is internal to this document, with an S prefix throughout.

S1. Appendix Roadmap

This appendix presents the per-generator and pipeline analyses, with each figure placed alongside the analysis that uses it. Fig. S4 is the full TemporalSpec pipeline diagram, and Fig. S2 is the audited cost-versus-OOD frontier; the rest visualize the per-generator results. The pooled detection and LOGO numbers behind these figures are tabulated in Table S1; what the figures add is the per-generator distribution behind those pooled numbers. Fig. S3 gives, for each generator, the detection AUC at $K=12$ (a), the LOGO in-distribution and unseen-generator AUC (b), and the dimensionality-matched FVMD PCA-13 LOGO bars on the matched 27,000-clip subset (c). Fig. S5 pairs the per-generator VBench human ranking with the ACF first-zero

ranking, summarized elsewhere by a single Spearman coefficient.

S2. Per-Generator Detection and LOGO Table

Table S1 reports the per-generator detection AUC and leave-one-generator-out (LOGO) in-distribution and out-of-distribution AUC behind the pooled headline.

S3. Full VBench Within-Prompt Concordance Table

Table S9 (plotted in Fig. S1) gives the full per-feature within-prompt concordance between MV features and VBench human pairwise preferences. It includes two rows omitted for space (sign-corrected acceleration kurtosis and MV variance, both under dynamic degree) and spells out both annotation labels (SMOOTH = motion smoothness, DYN. DEG. = dynamic degree). The sign-corrected concordance, 95% Wilson CIs, pair counts, and per-generator Spearman ρ ($n=3$) are computed on the same data throughout. The 1,071 pairs are the within-prompt generator pairs for which both a VBench human preference and a defined feature value exist; the smaller per-feature counts here (183 to 814) are the subsets that survive tie-handling and missing-feature exclusion for each feature. VBench supplies motion-smoothness and dynamic-degree annotations for its own generator set, which is not identical to the detection roster, so this concordance is a feature-validity check on whichever generators VBench annotates, not a per-detector comparison.

S4. Audited Metric Suite: Operating Points and Calibration

The audited metric tuple (operating-point TPR, partial AUC, and calibration) is computed under a single uniform L2-LR readout, the matched-harness default with no per-detector tuning, so that the comparison is controlled; the per-detector AUCs with tuned readouts are in the readout sweep (Sec. S11). All numbers are LOGO-OOD on the matched 27,000-clip subset, averaged over the 7 held-out generators. We define the metrics as follows. Partial AUC is the McClish-corrected pAUC over $FPR \leq 0.10$; TPR is read at the largest FPR not exceeding the target; and Brier score and ECE (10-bin) measure calibration, where lower is better. This section gives the precise metric definitions used throughout.

S5. Bootstrap 95% Confidence Intervals

Every AUC reported in this work carries a test-set bootstrap 95% confidence interval ($B=1000$ resamples with replacement, percentile method, seed=42). The model is fit once, under the exact splits used throughout this supplement, and

Table S1. **Detection (real vs. each generator) and cross-generator (LOGO) AUCs on GenVidBench Pair1 $\cup$ Pair2 ($K=12$).** *Left:* per-generator real-vs- G logistic regression on Tier 0 alone, Tier 1 alone, joint Tier 0+Tier 1, and a magnitude-free baseline. The pooled row uses all generators except T2VZ; the magnitude-free cell of that row is intentionally empty because the baseline is reported as a per-generator range only (0.81 to 0.93). *Right:* LOGO results. The ID column trains on real plus the held-out generator’s training fold; the OOD column trains on real plus the training folds of all other generators. Real pool = Vript $\cup$ HD-VG-130M ($n_{\text{real}}=26,956$). *Bottom block:* real-vs-real coherence probe. Generator codes: MS = ModelScope, VC2 = VideoCrafter2, SVD = Stable Video Diffusion, T2VZ = Text2Video-Zero.

Generator	n_{gen}	Detection AUC (real vs. generator)				LOGO		
		T0	T1	T0+T1	T0\mag+T1	ID AUC	OOD AUC	Δ
CogVideo	13853	0.966	0.798	0.971	0.881	0.972	0.942	−0.030
Mora	13667	0.840	0.778	0.856	0.812	0.859	0.750	−0.109
MS	13455	0.836	0.796	0.945	0.929	0.951	0.907	−0.044
MuseV	8374	0.854	0.794	0.888	0.886	0.888	0.652	−0.236
Pika	13450	0.806	0.637	0.859	0.858	0.857	0.761	−0.096
SVD	12822	0.886	0.800	0.896	0.832	0.890	0.831	−0.059
VC2	13489	0.861	0.808	0.913	0.868	0.910	0.891	−0.019
Pooled / Mean	89110	0.829	0.715	0.858	n/a	0.904	0.819	−0.085
<i>Real-vs-real probe (Vript vs. HD-VG-130M): T0 = 0.618 T1 = 0.579 T0+T1 = 0.628 T0\mag+T1 = 0.614</i>								

Table S2. **Headline bootstrap 95% CIs across methods.** ID detection: 80/20 stratified split. LOGO ID/OOD: mean across 7 held-out generators (per-generator CIs in Tables S3, S5). RvR: real-vs-real coherence probe (lower is better; chance = 0.5).

Method (27k subset)	ID detection	RvR	LOGO mean (ID / OOD)
MV T0+T1 (ours)	0.854 [0.843, 0.866]	0.643 [0.611, 0.675]	0.909 / 0.832
FVMD PCA-13	0.795 [0.782, 0.810]	0.552 [0.518, 0.583]	0.880 / 0.863
FVMD top-13 [LR coef]	0.754 [0.739, 0.768]	0.551 [0.519, 0.581]	0.836 / 0.761
FVMD full (1024d)	0.838 [0.825, 0.851]	0.579 [0.546, 0.613]	0.924 / 0.880
RAFT T0+T1	0.860 [0.847, 0.872]	0.627 [0.597, 0.658]	0.901 / 0.855
MV T0+T1 (full 116k)	0.858 [0.852, 0.864]	0.628 [0.612, 0.643]	0.904 / 0.819

only the predictions on the held-out test set are resampled; the interval therefore reflects sampling variability in the test fold, not the train fold. Table S2 gives the headline numbers for the five headline method/feature-set combinations, plus the full-dataset T0+T1 row: the LOGO mean across the 7 held-out generators, in-distribution detection (an 80/20 stratified split), and real-vs-real coherence.

S6. Distribution-Level FVMD per Generator

This section pairs the FVMD distribution-level score against our own per-generator detection AUC, to test whether the two methods agree on which generators are easy. For each generator, we compute the Fréchet distance between the real-pool activations (Vript and HD-VG-130M, $n=6,000$) and that generator’s activations ($n=3,000$) over the concatenated velocity and acceleration HOG descriptors (1024-d). We then place this distance beside the in-distribution AUC of the 13-d Tier 0+Tier 1 LR with the same generator held out (Table S7).

S7. Sensitivity to Feature-Extraction Hyperparameters

We sweep three orthogonal axes around the production configuration ($K=12$, Burg AR order $p_{\text{AR}}=4$, ACF max-lag $L=6$), holding the two non-swept axes at their defaults. Each configuration is evaluated on a fixed stratified 9,000-clip subset (1k per source across 9 sources: the seven generators plus the two real sources, Vript and HD-VG-130M; seed 42), using the same Tier 0+Tier 1 LR, median imputation, and standardization as the headline pipeline. The same subset drives the lattice sweep (Sec. S19), so the $K=12$ row below equals the lattice 4×4 row to within rounding. We report the LOGO mean across the seven held-out generators.

Sensitivity around the default operating point is small. As Table S14 shows, both the Burg AR order p_{AR} and the ACF max-lag L move LOGO ID and OOD by little (at most ~ 0.015 AUC) around the default. The non-I-frame block size K is the only axis with non-trivial movement: a coarser $K=8$ costs about two AUC points on ID and five on OOD, while a finer $K=16$ buys only a fraction of an AUC at the cost of discarding 45% of clips that lack sixteen non-I-frames. The setting $K=12$ is therefore the Pareto operating point.

The same $K=8$ axis independently confirms the structural T2VZ exclusion. We assembled a 30,000-clip cohort from the standard 27,000-clip subset plus 3,000 stratified T2VZ clips. Of these, 26,998 survive the $K=8$ non-I-frame filter, but none of the 3,000 T2VZ clips do; T2VZ is thus excluded by every $K\geq 8$ choice we have probed, not only at the headline $K=12$. The resulting nine-generator $K=8$ subset scores pooled ID 0.833, LOGO ID 0.888, LOGO OOD 0.794, and RvR 0.647. These track the 0.854/0.909/0.832/0.643 headline at $K=12$, confirming

that the $K=12$ cohort does not selectively benefit from a T2VZ-adjacent generator at the boundary. Finally, across the non-default configurations the per-generator MuseV OOD remains the worst case (as low as 0.60 at $K=8$), reinforcing that the MuseV failure mode is a property of the feature space and data coverage rather than an artifact of the sweep choices.

Table S3. **Per-generator LOGO bootstrap 95% CIs for the MV T0+T1 detector on the full 116k dataset.**

Held-out generator	ID AUC	OOD AUC
CogVideo	0.972 [0.968, 0.975]	0.943 [0.937, 0.948]
Mora	0.859 [0.851, 0.867]	0.749 [0.738, 0.760]
MS	0.951 [0.947, 0.955]	0.907 [0.900, 0.914]
MuseV	0.888 [0.881, 0.896]	0.653 [0.638, 0.666]
Pika	0.857 [0.849, 0.865]	0.761 [0.751, 0.771]
SVD	0.890 [0.883, 0.897]	0.831 [0.822, 0.840]
VC2	0.910 [0.904, 0.916]	0.891 [0.884, 0.897]

S8. GOP / Frame-Type Fingerprint Probe

We classify each frame of every clip in the 27,000-clip matched subset as I, P, or B by reading the per-block source field of the raw codec motion-vector dump: a frame is labeled I if the field is all-zero (no prediction), B if any block references a future frame (a positive reference, i.e. backward prediction), and P otherwise. We then run two probes: (i) a three-feature LOGO detector on the raw counts (n_I, n_P, n_B) , and (ii) a single-feature LOGO detector on n_{total} alone (a clip-length proxy).

The probe surfaces two complementary findings: one about clip-length leakage, and one about the canonical re-encode’s effect on frame-type structure.

Raw frame counts trivially leak the real-versus-generated distinction. A three-feature L2-LR on (n_I, n_P, n_B) , and a single-feature LR on n_{total} alone, both reach near-perfect cross-generator detection: LOGO mean ID and OOD are essentially indistinguishable from one. Table S4 shows why: generated clips are typically a few dozen frames whereas real clips run to several hundred, so clip length by itself is a near-perfect classifier. Reporting unaudited frame-count AUCs would therefore yield near-perfect detection driven entirely by this confound rather than by motion content.

The fractional GOP composition, by contrast, is approximately equalized by the canonical re-encode. In Table S4, the median within-clip ratios sit in narrow bands across real and generated sources for all three frame-type bins. Once the pipeline normalizes to the $K=12$ non-I-frame block, both the raw frame-count and the absolute GOP-bin signals are removed by construction; the residual 0.909 ID and 0.832 OOD signal must then come from within-block motion content, not from frame-type fingerprints. The $K=12$

choice therefore does two things: it quantifies the upper bound of clip-length leakage we are explicitly defending against, and it verifies that the canonical x264 re-encode equalizes the within-clip frame-type distribution that survives that normalization.

S9. Robustness to Post-Processing and Codec Variation

This section reports the full per-condition AUC for the robustness sweep over three perturbation families: H.264 CRF $\in \{23, 28, 32, 36\}$, frame-rate conversion from 8 to 24 fps, and spatial rescaling $\{0.5\times, 0.75\times\}$. The sweep runs on a 4,500-clip subset (500 per generator across 9 sources, including T2VZ), of which roughly 3,400 survive the $K=12$ filter on the canonical baseline. The canonical Tier 0+Tier 1 L2-LR is fit once on the clean 27,000-clip training subset and never refit on perturbed inputs; Table S6 then reports the per-condition pooled and per-generator AUCs.

The pooled canonical baseline in Table S6 (0.777) sits below the in-distribution detection headline (0.858) because the two numbers measure different subsets under the same pipeline. The headline is the in-distribution detection AUC on the full 116,066-clip evaluation set (Pair 1 and Pair 2 combined), where the real pool dominates ($n_{\text{real}}=26,956$). The robustness baseline, by contrast, is a deliberately balanced pooled AUC over 500 clips per source across the 9-source sweep, giving equal weight to the hardest per-generator subsets (Mora at 0.675, Pika at 0.656). Because the same canonical LR is scored on both subsets, the gap reflects subset composition, not pipeline inconsistency.

Three patterns are visible across the conditions in Table S6. First, the pooled AUC degrades monotonically with the H.264 quantizer, so motion-vector statistics carry a compression-quality bias but remain discriminative even at aggressive bitrates. Second, spatial rescaling is the gentlest perturbation in the sweep: the macroblock grid is itself a downsampled representation, so halving the input resolution shifts the feature distribution by less than a single CRF step. Third, frame-rate tripling from 8 to 24 fps is the largest single drop, as expected: the Tier 0+Tier 1 features summarize displacement statistics over a $K=12$ non-I-frame window, and re-encoding at three times the source frame rate compresses each clip’s motion into a temporally finer grid that no longer matches the training distribution. None of the conditions are catastrophic, however: every degraded subset remains well above chance and within 0.07 AUC of the canonical pipeline.

Mitigation for 8 to 24fps drift. The frame-rate condition is the only one in the sweep that drops AUC by

Table S4. **Per-generator frame-type distribution after canonical re-encode.** Counts are raw frame counts per clip (mean), ratios are within-clip ratios (median). The *fractional* GOP structure (r_I, r_P, r_B) is approximately equalized across real and generated sources by the canonical x264 re-encode pipeline; the *raw* totals trivially separate them because generated clips are far shorter than real clips.

Source	real?	n	$\bar{n}_I$	$\bar{n}_P$	$\bar{n}_B$	$\tilde{r}_I$	$\tilde{r}_P$	$\tilde{r}_B$
HD-VG-130M	1	3000	8.35	269.3	490.9	0.009	0.327	0.660
Vript	1	3000	2.85	106.3	213.1	0.008	0.309	0.680
CogVideo	0	3000	1.34	10.3	21.4	0.030	0.242	0.727
Mora	0	3000	1.08	6.9	16.1	0.042	0.250	0.708
MS	0	3000	1.32	4.9	9.8	0.062	0.250	0.688
MuseV	0	3000	1.00	3.4	8.6	0.077	0.231	0.692
Pika	0	3000	1.12	19.4	39.0	0.014	0.278	0.694
SVD	0	3000	1.11	4.4	8.5	0.071	0.286	0.643
VC2	0	3000	1.16	4.2	10.6	0.062	0.250	0.688

Table S5. **Per-generator LOGO bootstrap 95% CIs for RAFT T0+T1 on the 27k matched subset.**

Held-out generator	ID AUC	OOD AUC
CogVideo	0.922 [0.908, 0.935]	0.802 [0.779, 0.823]
Mora	0.880 [0.864, 0.895]	0.874 [0.857, 0.891]
MS	0.824 [0.805, 0.843]	0.782 [0.759, 0.804]
MuseV	0.959 [0.950, 0.967]	0.876 [0.858, 0.893]
Pika	0.930 [0.919, 0.941]	0.883 [0.868, 0.898]
SVD	0.867 [0.851, 0.884]	0.861 [0.842, 0.878]
VC2	0.923 [0.911, 0.935]	0.908 [0.894, 0.920]

Table S6. **Robustness sweep.** Pooled ID AUC with bootstrap 95% CIs ($B=1000$, percentile, seed 42) and Δ versus the canonical baseline (H.264 CRF = 23). MuseV and SVD have empty per-generator subsets because their 4-frame canonical clips fail the $K=12$ non-I-frame filter after re-encoding. The HEVC condition is omitted from the LR-perturbation sweep because the detector is fit on H.264 features; an empirical probe of the extractor itself (below) confirms that the FFMpeg MV extractor returns motion vectors for HEVC bitstreams on FFMpeg $\geq 7.1.1$.

Condition	n	Pooled AUC [95% CI]	Δ
H.264 CRF 23 (baseline)	3405	0.777 [0.759, 0.795]	0
H.264 CRF 28	3411	0.760 [0.741, 0.778]	-0.018
H.264 CRF 32	3374	0.738 [0.720, 0.758]	-0.039
H.264 CRF 36	3340	0.728 [0.708, 0.747]	-0.049
Spatial rescale 0.75×	3372	0.764 [0.745, 0.781]	-0.014
Spatial rescale 0.5×	3215	0.746 [0.728, 0.765]	-0.031
Frame-rate 8 to 24 fps	4403	0.710 [0.691, 0.729]	-0.067

Per-generator AUC (5 generators with valid $K=12$ post-re-encode)

CogVideo: baseline 0.901; range [0.843 (CRF 36), 0.901]; fr. 24 fps 0.535.

Mora: baseline 0.675; range [0.544 (CRF 36), 0.728 (fr. 24 fps)].

MS: baseline 0.859; range [0.802 (rescale 0.5×), 0.866 (CRF 28)].

Pika: baseline 0.656; range [0.656, 0.702 (CRF 36)].

VC2: baseline 0.795; range [0.712 (CRF 36), 0.795].

more than 0.04, so we probe the natural mitigation: re-applying the canonical 8 fps re-encode, before MV extraction, to a clip that arrived perturbed at 24 fps. Concretely, we take the same 4,500-clip subset, apply the 24 fps perturbation that produced the 0.710 row, and pass

each 24 fps re-encode through a second canonical 8 fps re-encode (libx264, CRF= 23, GOP= 12, main profile, 8 fps, 512×512) before MV extraction. The downstream pipeline ($K=12$ filter, Tier 0+Tier 1, canonical LR) is unchanged, and we never refit on the perturbed inputs. Pooled AUC recovers to 0.7625 [0.7440, 0.7815] from the 0.710 baseline, a +0.052 gain that restores most of the original -0.067 drop. The recovery is not free. The two-stage temporal downsampling reduces the effective frame budget, so 1139 of 4500 clips fall below the $K=12$ non-I-frame minimum and are dropped: MuseV and SVD lose all 500 clips, because their canonical-baseline survivors were already at the edge of the filter, while VC2 loses about a quarter of its clips (376 of 500 survive) and CogVideo, Mora, MS, and Pika each lose $\leq 1\%$. The mitigation is therefore practical when clips have enough frames to absorb the $3\times$ temporal subsampling; it is a deployment-time pipeline choice, not a model change. A no-dropout alternative that operates on the bitstream without a second re-encode (frame-skipping during MV extraction at 24 fps) is left to future work.

Empirical FFMpeg HEVC behavior. We verified our extractor’s codec coverage on the system FFMpeg 7.1.1 build used throughout this work, which is configured with both H.264 and HEVC support. For each of three deterministically sampled clips from the parent subset, we re-encoded the source as both H.264 (CRF= 23, GOP= 12, 8 fps) and HEVC (same settings), ran the FFMpeg MV extractor on each output, and recorded the resulting motion-vector array shape. All three HEVC re-encodes returned MV arrays of the same shape as their H.264 counterparts ($(T, H_{MB}, W_{MB}, 5)$ with $T \geq 12$; specifically (26, 40, 68, 5), (33, 30, 30, 5), and (13, 44, 80, 5)), confirming that FFMpeg $\geq 7.1.1$ populates the codec MV side-data channel for HEVC inputs as well as H.264. This is a behavioral change from older builds, where the channel was H.264-only. We therefore do not treat HEVC input as an

extractor-level limitation: every clip in this paper is normalized through the canonical H.264 re-encode before MV extraction, so the detector’s training distribution is single-encoder by design, not by tool restriction. A robustness number against unnormalized HEVC source clips would require refitting the detector on HEVC features, and is left to future work; AV1 was not probed and remains future work for the same reason.

Table S7. **Distribution-level FVMD per generator vs. MV ID detection AUC.** “combined FVMD” is the Fréchet distance between real-pool and per-generator activations over the concatenated velocity+acceleration HOG (1024-d). MV ID AUC is the in-distribution AUC of the T0+T1 LR with this generator as the held-out test set. Spearman $\hat{\rho} = -0.714$ ($p=0.071$, $n=7$; paired bootstrap 95% CI $[-1.00, +0.06]$, $B=10,000$, seed 42; $\Pr(\hat{\rho}<0)=0.97$): higher FVMD (i.e., farther from real) does *not* imply higher MV-AUC (i.e., easier MV detection); the two methods tend to rank generators inversely (suggestive at $n=7$), consistent with complementary motion descriptors.

Generator	combined FVMD	MV ID AUC
MuseV	14738.7	0.888
Pika	8711.0	0.857
Mora	6983.5	0.859
VC2	6750.3	0.910
CogVideo	6104.7	0.972
SVD	5591.2	0.890
MS	4274.7	0.951

S10. MuseV Deep-Dive: per-Feature Cohen’s d and LOO Ablation

Following the cross-generator failure of MuseV (LOGO OOD 0.652 on the full 116,066-clip dataset and 0.673 on the matched 27,000-clip subset), we run two complementary analyses to locate the feature responsible.

A per-feature effect-size analysis between MuseV ($n=3,000$) and the pooled non-MuseV generated clips ($n=18,000$) identifies four features with $|d| \geq 1$, all listed in Table S8. MuseV occupies a region of motion space with higher acceleration kurtosis, higher spectral flatness, more zero-motion macroblocks, and lower mean motion magnitude than the rest of the training pool.

Table S8. **Per-feature Cohen’s d between MuseV and the pooled non-MuseV generated clips.** Only features with $|d| \geq 1$ are shown. Positive d means MuseV has the higher feature mean.

Feature	Cohen’s d
median acceleration kurtosis	+2.01
median spectral flatness	+1.35
MV sparsity	+1.03
mean MV magnitude	−1.01

Table S9. **Full within-prompt agreement table.** Sign-corrected concordance (chance = 0.500), 95% Wilson CIs, number of pairs, and per-generator Spearman ρ ($n=3$). **Bold** = best in family. Both Tier-0 (magnitude) and Tier-1 (spectral) features are crossed against both annotations.

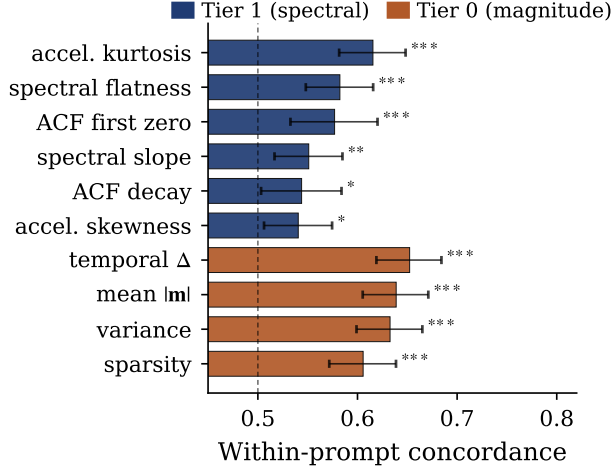
Annot.	Feature	Concord.	95% CI	Pairs	ρ_{gen}
<i>Motion-smoothness annotation</i>					
SMOOTH	spectral slope	0.551	[0.517, 0.585]	813	+1.000
SMOOTH	spectral flatness	0.582	[0.548, 0.616]	814	+0.500
SMOOTH	ACF first-zero	0.577	[0.533, 0.620]	487	+1.000
SMOOTH	accel. kurtosis	0.615	[0.582, 0.648]	814	+0.500
SMOOTH	mean MV magnitude	0.361	[0.329, 0.395]	814	−0.500
SMOOTH	MV temporal diff	0.348	[0.316, 0.381]	814	−0.500
<i>Dynamic-degree annotation</i>					
DYN. DEG.	spectral slope	0.521	[0.469, 0.573]	351	−1.000
DYN. DEG.	ACF first-zero	0.525	[0.452, 0.596]	183	−1.000
DYN. DEG.	accel. kurtosis (sign-corr.)	0.578	[0.526, 0.629]	351	−0.500
DYN. DEG.	mean MV magnitude	0.587	[0.535, 0.637]	351	+0.500
DYN. DEG.	MV variance	0.593	[0.540, 0.643]	351	+0.500
DYN. DEG.	MV temporal diff	0.595	[0.543, 0.645]	351	+0.500

A leave-one-feature-out ablation under LOGO, with MuseV held out, localizes the failure to the single feature with the largest MuseV distribution shift. Removing the median acceleration kurtosis is the single most impactful deletion, a +5.8-point MuseV OOD gain consistent with the model overfitting a MuseV-incompatible kurtosis statistic learned from the other six generators. Removing any of median acceleration skewness, MV sparsity, or mean MV magnitude instead costs roughly five AUC points each, confirming that the remaining Tier 1 features carry generalizable signal.

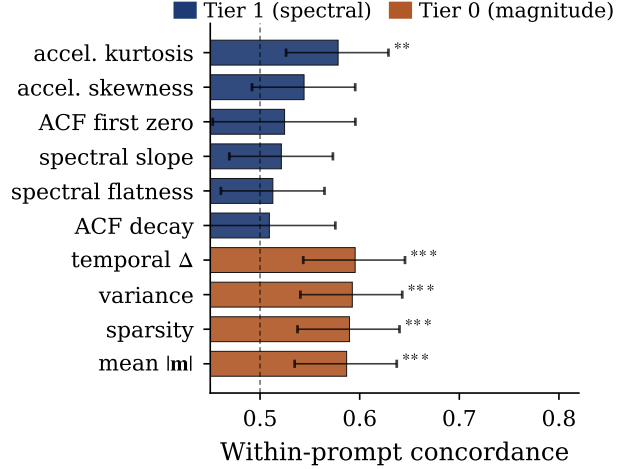
S11. Readout Sweep: Features Across Backbones

We keep L2-LR as the headline classifier so that the comparison with FVMD and RAFT stays linear-equivalent. But as a feature-engineering contribution, the 13-d Tier 0+Tier 1 vector should carry signal independently of the downstream classifier head. To test this, we hold the impute-then-standardize pipeline fixed and swap the readout across seven families: L2 logistic regression; gradient-boosted trees (XGBoost and LightGBM, both with $n_{\text{est}}=100$, depth 5, learning rate 0.1); a random forest ($n_{\text{est}}=200$, depth 10); an RBF-kernel SVM ($C=1$); and two multi-layer perceptrons, with two and four hidden layers respectively (ReLU activations, Adam optimizer, early stopping with a 10% validation hold-out). All readouts share the same seed 42, the same 27,000-clip subset, and the same LOGO/RvR protocol. Table S10 reports the numbers ranked by LOGO OOD, listing each classifier alongside its training wall-clock to make the cost trade-off explicit.

Two results stand out. First, all seven readouts land in a 0.820 to 0.849 LOGO OOD band: a 2.9-point spread across linear, tree-ensemble, kernel, and neural-network



(a) Motion smoothness.



(b) Dynamic degree.

Figure S1. **Within-prompt agreement of MV features with human preferences.** Sign-corrected concordance with 95% Wilson intervals; chance (0.5) dashed. Asterisks: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. Tier 1 features (blue) exceed chance on smoothness while the Tier 0 magnitude features (orange) fall below it; on dynamic degree, signal appears mainly in the Tier 0 magnitude features.

Table S10. **Readout sweep on the 13-feature Tier 0+Tier 1 vector, 27,000-clip subset.** Same impute-then-standardize pipeline, same LOGO/RvR protocol, same seed; only the classifier head changes. Lower RvR is better. Time is training wall-clock for the full LOGO+RvR harness on 8 CPU cores; inference for any of these readouts is sub-millisecond per clip.

Readout	ID	LOGO ID	LOGO OOD	RvR	Time
L2-LR (headline)	0.854	0.909	0.832	0.643	0.9 s
XGBoost	0.898	0.944	0.845	0.625	3.6 s
LightGBM	0.897	0.944	0.845	0.635	11 s
Random forest	0.877	0.929	0.820	0.634	14 s
SVM-RBF	0.886	0.931	0.826	0.639	383 s
MLP (2 hidden)	0.886	0.933	0.845	0.625	20 s
MLP (4 hidden)	0.890	0.936	0.849	0.627	34 s

classifiers, which shows that the 13-d feature set is the signal carrier, not the LR readout. Second, the four-hidden-layer MLP reaches the strongest LOGO OOD of the sweep, 0.849, lifting the headline L2-LR by 1.7 points on cross-generator generalization at the cost of a 34-second fit; the codec-MV features are thus compatible with neural readouts as well as linear ones. We retain L2-LR as the headline readout both for linear-equivalent comparison with FVMD and RAFT, and because its single-line decision boundary makes the feature-attribution audit in Sec. S10 interpretable.

S12. D3 Head-to-Head on the Matched 27,000-Clip Subset

We evaluate D3 [49] in two settings on the same 27,000-clip subset (3,000 per source across 9 sources) used for MV,

FVMD, and RAFT. Both settings use per-frame XCLIP-ViT-B/16 embeddings and the published second-order temporal central difference $\Delta^2 f(t) = f(t+1) - 2f(t) + f(t-1)$. The first time-mean-pools the D3 features into the 768-d trajectory and fits it through the same L2-LR harness as MV, FVMD, and RAFT. The second uses D3 as published and training-free, taking $\text{score} = \text{std}_t \Delta^2 f(t)$ as the per-clip scalar; because no classifier is fit, ID and OOD coincide by construction. Table S12 reports both settings alongside our 13-d MV detector on the same subset.

The contrast between the two D3 rows is the headline finding: D3’s published head is the contribution. Substituting our learned LR readout costs more than twenty AUC points on cross-generator OOD, which suggests that the 768-d XCLIP embedding does not carry detector-relevant content in a form a linear L2-LR can recover from 21,600 training rows. Relative to our 13-d MV detector, the D3 native head ties on pooled ID, trails by roughly two points

on LOGO ID, and gains five to six points on LOGO OOD. It is also closer to chance on the real-versus-real probe, so its dataset-bias spoofability is lower than ours. That cross-generator gain comes at the cost of an XCLIP-ViT-B/16 forward pass per clip on a GPU, whereas the codec-MV detector ties on pooled ID at the same matched subset with a CPU-only 14 ms readout; the two are therefore different operating points on the cost-versus-ODD frontier, not a strict accuracy ranking.

S13. Deployment Guidance and Reproducibility

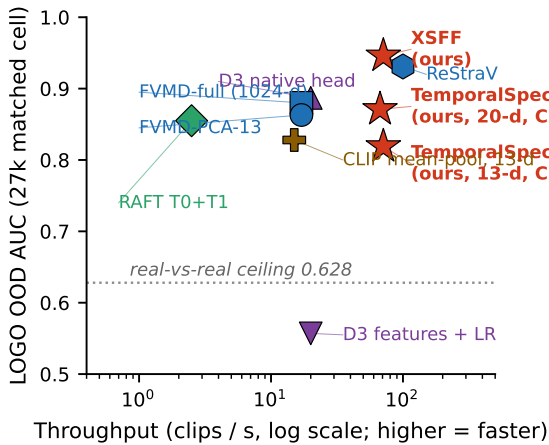


Figure S2. **Audited cost-vs-ODD frontier on the matched 27,000-clip subset.** Audited LOGO OOD vs. per-clip post-decode cost (measured for TemporalSpec, RAFT, WaveRep; order-of-magnitude estimated for FVMD, D3, ReStraV, CLIP, grayed). Red stars mark our case studies: **TemporalSpec** (CPU, 13-d, 0.832), **TemporalSpec+aug** (CPU, 20-d, 0.871), and **XSFF** (MV + ReStraV, 0.946, one ViT-S/14 GPU forward). WaveRep [9] is the strongest re-evaluated detector (0.996, chance RvR, certified); ReStraV [17] follows at 0.931. Dashed: Pareto frontier; dotted: the 0.643 dataset-identity floor. The CPU 0.871 and GPU 0.946 are distinct operating points, not a CPU-only claim.

We recommend the codec-MV detector as one channel of evidence in an ensemble, not as a stand-alone gate. For adversarial deployment, we recommend the 10-feature magnitude-free baseline (T0\mag+T1), which removes the partial dataset-bias signal carried by magnitude features. The detector is calibrated for H.264 inputs and should not be over-extended to codec families it has not been re-fit on; cross-codec input that bypasses the canonical re-encode is left to future work. Finally, because a single unseen generator can drop OOD sharply (MuseV holds out at LOGO OOD 0.673, about 19 points below the 0.866 mean of the other six held-out generators on the matched subset), we recommend ensemble deployment with at least one

appearance-domain channel (such as D3’s native head) for any setting in which the generator roster is open-ended.

For reproducibility, the following artifacts are available at <https://github.com/KurbanIntelligenceLab/vidaudit>:

- x264 canonical re-encode driver script (FFmpeg 7.1.1, exact command lines).
- MV extraction Python wrapper (binding to FFmpeg’s MV side-data channel).
- 13-d Tier 0+Tier 1 feature computation module + the 7-d augmented vector-field feature module (Sec. S24).
- Trained L2-LR weights for the 13-d headline detector at seed 42 (.pkl), plus gradient-boosted readout weights for the 20-d augmented configuration.
- Matched 27,000-clip CSV with stratified per-generator splits.
- Full 116,066-clip Pair1UPair2 index.
- AIGVDBench within-LOGO retrained classifier weights (random-forest readout, Sec. S26).
- ReStraV reproduction driver (DINOv2 ViT-S/14 feature extraction at the matched harness, Sec. S25; the original repository ships the native MLP head and we add the matched-harness wrapper).
- Multi-seed reproducibility scripts (seeds {0, 7, 17, 42, 99}, Sec. S21).
- Robustness sweep + FPS-mitigation pipelines (Sec. S9).
- A re-scoring entry point, `rescore_detector.py`, that ingests detector predictions on the audited subset and returns audited LOGO and RvR scores, so any detector can be placed on the audited leaderboard without re-running the full harness.

All evaluations are deterministic at seed 42: the 80/20 stratified split, the LOGO assignment, and the bootstrap resampling all use the same seed. The canonical H.264 pipeline is CRF= 23, GOP= 12, main profile, 8 fps, 512×512 bilinear-square rescale, FFmpeg 7.1.1. The L2-regularized logistic-regression readout (the matched-harness head used throughout the paper) is fit on Tier 0+Tier 1 features with median imputation, per-feature z -scoring, balanced class weights, and L2 regularization strength $C=1$ for the headline rows; the “inner CV” variant instead selects C by 5-fold inner cross-validation over $C \in \{0.01, 0.1, 1, 10\}$ (see Sec. S14). The multi-seed grid is {0, 7, 17, 42, 99} (Sec. S21). We do not redistribute the underlying GenVidBench or AIGVDBench videos; reproduction requires obtaining them from their original sources, after which the released harness regenerates every subset.

S14. Matched-Harness Setup for FVMD, RAFT, and D3 Comparison

The matched 27,000-clip stratified subset comprises 3,000 clips per generator across 7 generators (T2VZ excluded by

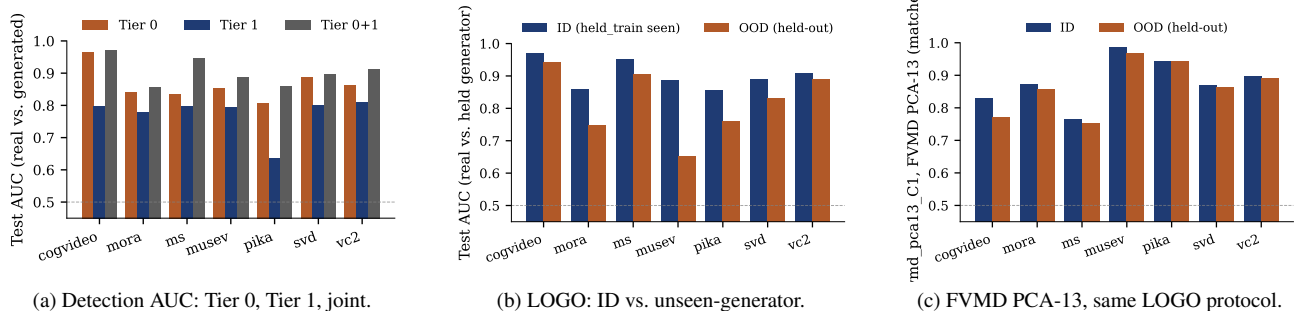


Figure S3. **Per-generator AUC under leakage-controlled $K=12$ extraction.** (a) Tier 1 spectral features add a consistent margin over Tier 0 alone. (b) Six of seven generators retain OOD AUC ≥ 0.75 ; MuseV is the failure mode of the current feature space. (c) Dimensionality-matched FVMD retains 0.863 mean OOD AUC across the 7 held-out generators, 3.1 points above our MV detector at the same dimensionality on this matched subset, but with a substantially larger raw feature extractor (a pretrained PIPs++ point-tracker backbone). Numeric values are in Table S1.

Table S11. **Per-stage CPU timing on 200 clips** (AMD EPYC 7742, single core, FFmpeg 7.1.1). Sources arriving in canonical H.264 skip the first stage; sources arriving in foreign codecs incur it once per clip as a pre-processing pass.

Stage	mean (ms)	median	p95	p99
Canonical H.264 re-encode (one-off)	4555.6	2464.0	14599.0	39715.9
MV decode (FFmpeg + extract_mvs)	2686.3	1669.4	7926.8	19212.5
Feature computation	14.2	21.7	26.6	41.5
End-to-end (foreign codec input)	7256.1	4308.1	20048.9	53442.7
End-to-end (canonical-H.264 input)	2700.5	1691.1	7953.4	19254.0

the $K=12$ filter) plus 3,000 each from Vript and HD-VG-130M. Beyond the published full 1024-d FVMD, we report two dimensionality-matched FVMD variants on this subset. FVMD PCA-13 fits a PCA on the 21,600-row training fold of each LOGO split and projects all clips to 13 components before LR; FVMD top-13 [LR coef] instead selects the 13 features with the largest L2-LR coefficient magnitude on the same training fold. Both variants wrap an outer estimator with a fold-aware feature-selection step so that the selection is never trained on the test fold. The four FVMD configurations on this subset (LOGO ID / OOD / RvR) are: full 1024-d 0.924/0.880/0.574; full with 5-fold inner CV 0.942/0.892/0.592; PCA-13 0.880/0.863/0.555; top-13 0.836/0.761/0.548. Our headline comparison uses the full and PCA-13 rows (the headline metric and the dimensionality-matched comparison to our 13-d detector); the inner-CV and top-13 variants are the regularization and feature-selection ablations.

The RAFT comparison decodes 12 frames at 256×256 , runs pretrained RAFT-Large (C+T+SKHT+V2 preset, 5.26M parameters) on the 11 consecutive frame pairs, and average-pools the resulting dense flow to the same 32×32 MV-macroblock lattice. The pooled $(\Delta x, \Delta y)$ field is then fed to the exact Tier 0+Tier 1 feature pipeline used for codec MVs, so any difference between the MV and RAFT rows of the matched-harness comparison reflects the motion

source alone (codec block-matching versus learned dense flow) rather than the downstream features. To bracket regularization, every method is additionally reported with 5-fold inner cross-validation over $C \in \{0.01, 0.1, 1, 10\}$ alongside the headline $C=1$ row.

A per-method extraction-failure audit on the 27,000-clip subset records zero failed clips for any of FVMD, RAFT, or MV. The same MV pipeline re-run on this subset reproduces our headline (LOGO ID/OOD 0.909/0.832, RvR 0.643), within 0.02 AUC of the full 116k-clip result (0.904/0.819/0.628), so the cross-method comparison numbers sit on essentially the same operating point as the GenVidBench-wide ones.

Pipeline parity across baselines. Every matched-harness detector row (FVMD, RAFT, D3, CLIP, and WaveRep) consumes the same canonical H.264 re-encode (control C1 of the six-control audit; CRF=23, GOP=12, main profile, 8 fps, 512×512). The $K=12$ non-I-frame filter (control C2, the leakage audit) is motion-vector-specific: it applies to the codec-MV and RAFT rows, whereas the frame-based detectors (D3, CLIP, and WaveRep) score the re-encoded frames directly and so are not exposed to the MV frame-count channel C2 removes. The baselines thus differ in what they compute on the same re-encoded clips (point-tracker velocity/acceleration histograms for FVMD, RAFT-Large dense flow at 256×256 for RAFT, XCLIP-ViT-B/16 embeddings for D3, CLIP-ViT-B/32 mean-pool for the appearance baseline, and DINOv2 ViT-B/14 forensic LLR for WaveRep), not in which clips they see. The full 116k-row training subset likewise applies the same canonical re-encode; the 27,000-clip subset is the stratified sub-selection used for cross-method comparison.

Why DeMamba is not re-scored. We attempted to add DeMamba [8] as a fourth published detector. It releases

Table S12. **D3 head-to-head on the matched 27,000-clip subset.** The native published head returns a single scalar, so its ID and OOD AUCs coincide. Pooled rows are pooled across the entire matched subset; LOGO rows are means across the seven held-out generators. Per-generator OOD range is reported only where it differs from the mean.

Method	Pooled ID	LOGO ID	LOGO OOD	RvR	OOD range
MV T0+T1 (ours)	0.854	0.909	0.832	0.643	n/a
D3 features + L2-LR	0.593	0.668	0.557	0.546	[0.467, 0.590]
D3 native head (no refit)	0.870	0.887	0.887	0.421	[0.787, 0.972]

training-only code but withholds its pretrained checkpoints (the authors cite a corporate open-source policy on the public issue tracker) and ships no inference path, with data and weight loading hardwired to a private bucket. Re-scoring it under our harness would require retraining from scratch on its million-scale GenVideo training set, which we do not mirror and which would not constitute the published detector. We therefore exclude it under the same no-public-weights rationale as the code-less families above, rather than report a non-comparable retrained number.

Table S13. **Lattice resolution sweep on the shared 9,000-clip subset** (seed 42; the 4×4 row equals the sensitivity $K=12$ row to within rounding). The 2×2 lattice underperforms; 4×4 and 8×8 are tied within bootstrap noise on all three axes, so we adopt the cheaper 4×4 (a quarter the cells).

Lattice	LOGO ID	LOGO OOD	RvR (lower better)
2×2 (4 cells)	0.880	0.800	0.601
4×4 (16 cells, default)	0.904	0.825	0.607
8×8 (64 cells)	0.907	0.826	0.607

S15. Per-Feature Properties and Tier 0 / Tier 1 Orthogonality

Table S14. **Sensitivity sweep over K, p, L** on the shared 9,000-clip subset (seed 42). Default config ($K=12, p=4, L=6$) is the bolded row. $K=16$ reduces n to 4,994 because many clips have fewer than 16 non-I-frames.

Config	LOGO ID	LOGO OOD	gap	n_{rows}
$K=8$	0.886	0.771	+0.115	9000
$K=12$ (default)	0.903	0.825	+0.078	9000
$K=16$	0.904	0.831	+0.073	4994
$p=2$	0.908	0.826	+0.081	9000
$p=6$	0.897	0.811	+0.086	9000
$L=4$	0.913	0.836	+0.078	9000
$L=8$	0.902	0.821	+0.081	9000

This section gives, for each of the 13 features in the headline detector, its definition, range, scaling behavior, and what bias it captures. Table S15 summarizes; the per-feature notes that follow give the derivation. The result is the construction-level Tier 0/Tier 1 orthogonality: every

Tier 1 feature is exactly invariant under positive rescaling of the underlying motion-magnitude time series, while every Tier 0 feature carries only that scaling information.

* Sparsity is invariant under positive rescaling because $1[m=0]$ is preserved by multiplication by $c > 0$. Sparsity is *not* invariant if the codec dead-zone threshold δ is rescaled; in our pipeline δ is fixed by the encoder, so the property holds in practice.

† Temporal-difference is invariant under cyclic shift; under acyclic shift with r_{-1} undefined the first term is dropped, an $O(1/T)$ effect.

‡ Spectral-slope is shift-invariant in the sense that the slope of $\log S_r$ under cyclic shift is unchanged (Wiener-Khinchin).

Scaling-invariance of Tier 1. The construction-level invariance follows because Tier 1 features all factor through quantities that are quadratic in r (PSD, autocorrelation, second-order moment of acceleration). Under $r'(t) = c \cdot r(t)$ with $c > 0$:

- PSD: $S_{r'}(f) = c^2 S_r(f)$, so the log-log fit $\log S(f) = C - \beta \log f$ becomes

$$\log S_{r'}(f) = (2 \log c + C) - \beta \log f, \quad (\text{S15.1})$$

with slope β unchanged and only intercept C shifted. Spectral slope features invariant.

- Wiener entropy (spectral flatness):

$$\begin{aligned} \frac{\exp(\text{mean}_f \log S_{r'})}{\text{mean}_f S_{r'}} &= \frac{c^2 \exp(\text{mean}_f \log S_r)}{c^2 \text{mean}_f S_r} \\ &= \frac{\exp(\text{mean}_f \log S_r)}{\text{mean}_f S_r}, \end{aligned} \quad (\text{S15.2})$$

so c^2 cancels. Flatness features invariant.

- Autocorrelation:

$$R_{r'}(\tau) = \frac{c^2 \text{Cov}_r(t, t+\tau)}{c^2 \text{Var}(r)} = R_r(\tau), \quad (\text{S15.3})$$

so c^2 cancels in both numerator and denominator. Decay time τ_0 and first-zero lag invariant.

- Acceleration moments: with $a'_t = c \cdot a_t$,

$$\begin{aligned} \frac{\mathbb{E}[a'^4]}{(\mathbb{E}[a'^2])^2} &= \frac{c^4 \mathbb{E}[a^4]}{(c^2 \mathbb{E}[a^2])^2} \\ &= \frac{\mathbb{E}[a^4]}{(\mathbb{E}[a^2])^2}, \end{aligned} \quad (\text{S15.4})$$

Table S15. **Mathematical properties of the 13 Tier 0 + Tier 1 features.** Feature IDs: the Tier 0 rows are T0-1 to T0-4 in order, and the Tier 1 base statistics are T1-1 (spectral slope), T1-2 (spectral flatness), T1-3 (ACF decay time), T1-4 (ACF first-zero), T1-5 (acceleration kurtosis), and T1-6 (acceleration skewness), with T1-1 to T1-3 each contributing a median and an IQR row. Symbols: $r(t)$ is the per-cell mean MV-magnitude time series of a clip with T samples; $S_r(f)$ is its PSD; $R(\tau)$ its autocorrelation; $a_t = r_t - 2r_{t-1} + r_{t-2}$ the acceleration. Scaling: behavior under $r'(t) = c \cdot r(t)$ for $c > 0$. Temporal-shift: behavior under $r'(t) = r(t - \tau_0)$ with cyclic boundary. “Bias captured” = the directional prediction from Sec. S16 for the generated-vs-real contrast.

Feature	Definition	Range	Scaling under $r \rightarrow cr$	Temp. shift	Tier	Bias predicted
<i>Tier 0 (magnitude statistics):</i>						
mean magnitude	$\frac{1}{T} \sum_t \bar{m}_t$	$[0, \infty)$	$\propto c$	invariant	T0	gen<real (under-motion)
sparsity	$\frac{1}{TN_{MB}} \sum_t \sum_{ij} \mathbf{1}[m_t^{(i,j)}=0]$	$[0, 1]$	invariant*	invariant	T0	gen>real (static-content)
magnitude variance	$\text{Var}_t(\bar{m}_t)$	$[0, \infty)$	$\propto c^2$	invariant	T0	gen<real (less variability)
temporal difference	$\frac{1}{T-1} \sum_t \bar{m}_t - \bar{m}_{t-1} $	$[0, \infty)$	$\propto c$	invariant†	T0	gen<real (smoother)
<i>Tier 1 (spectral, ACF, acceleration shape):</i>						
spectral slope (median)	median across cells of β s.t. $\log S_r(f) = C - \beta \log f$	$\mathbb{R}$	invariant	invariant‡	T1	gen>real (more low- f mass)
spectral slope (IQR)	IQR of the same	$[0, \infty)$	invariant	invariant	T1	n/a (spread, not direction)
spectral flatness (median)	median of $\frac{\exp(\text{mean}_f \log S_r(f))}{\text{mean}_f S_r(f)}$	$[0, 1]$	invariant	invariant	T1	gen<real (less broadband)
spectral flatness (IQR)	IQR of the same	$[0, 1]$	invariant	invariant	T1	n/a (spread)
ACF decay time (median)	median across cells of τ_0 s.t. $R(\tau) \approx \exp(-\tau/\tau_0)$	$[0, \infty)$	invariant	invariant	T1	gen>real (longer decay)
ACF decay time (IQR)	IQR of the same	$[0, \infty)$	invariant	invariant	T1	n/a (spread)
ACF first-zero (median)	median of first $\tau > 0$ s.t. $R(\tau)=0$	$[1, T]$	invariant	invariant	T1	gen>real (later zero)
acceleration kurtosis (median)	median of $\frac{\mathbb{E}[a^4]}{(\mathbb{E}[a^2])^2} - 3$	$[-2, \infty)$	invariant	invariant	T1	gen<real (fewer jerky spikes)
acceleration skewness (median)	median of $\frac{\mathbb{E}[a^3]}{(\mathbb{E}[a^2])^{3/2}}$	$\mathbb{R}$	invariant	invariant	T1	n/a (sign-asymmetry, no directional prediction)

so c^4 cancels. Kurtosis (and skewness, with c^3/c^3 cancellation) invariant.

Conversely, the Tier 0 features carry the scaling information that Tier 1 throws away: mean magnitude scales linearly, magnitude variance scales quadratically, temporal difference scales linearly, and sparsity is the only Tier 0 feature that is also scale-invariant (it counts zero macroblocks rather than measuring their magnitude). The two tiers therefore measure orthogonal axes of the same underlying $r(t)$, which is the analytic explanation for the $\rho=\pm 1.0$ ranking pattern observed empirically against VBench’s smoothness and dynamic-degree annotations (Sec. S3).

Why each Tier 1 feature should differ for generated clips. The bias-prediction column of Table S15 follows directly from the spectral-bias prediction derived in Sec. S16 and the four signed signatures P1–P4 stated there. In short: generator-side temporal high-frequency suppression shifts spectral mass to low f , which (a) steepens the log-log slope β , (b) makes the spectrum less uniform (lower flatness), (c) broadens the autocorrelation (longer τ_0 , later first-zero), and (d) suppresses the heavy-tailed jerky-event spikes that drive acceleration kurtosis. The skewness prediction is sign-asymmetric and has no directional prior. Tier 0 captures complementary content: overall motion amplitude rather than temporal structure. Its directional prediction is that generated content has, on average, lower motion magnitude (gen<real), less temporal variability (gen<real), smoother frame-to-frame changes (gen<real), and more zero-motion macroblocks (gen>real). This is consistent both with the architectural bias (a temporal bandwidth limit

suppresses small-magnitude flicker) and with the empirical observation that generated clips often depict less motion overall.

S16. Why Codec MV Spectra Should Differ Between Real and Generated Video

This section gives the derivation behind the spectral-bias prediction and argues why the codec-MV substrate is theoretically *orthogonal* to the deep-feature trajectories that power D3 and ReStrAV.

Step 1: the codec MV operator. The MV field is the image of the true flow $\mathbf{u}_t$ under $\Phi = \mathcal{S}_\lambda \circ \mathcal{Q}_\delta \circ \mathcal{L}_M$. Each factor has a precise spectral or magnitude-domain action: $\mathcal{L}_M$ multiplies $\hat{\mathbf{u}}$ by a sinc envelope with first zero at $1/M$ cycles/pixel; $\mathcal{Q}_\delta$ zeroes magnitudes below the dead-zone δ and rounds the rest to the integer lattice; $\mathcal{S}_\lambda$ minimizes $D_{\text{SAD}} + \lambda R$ over a neighborhood of candidates. Φ thus retains components of $\mathbf{u}_t$ with spatial frequency $\leq 1/M$ and magnitude $\geq \delta$, and regularizes the residual toward temporal smoothness.

Step 2: generator-side temporal high-frequency suppression. The empirical literature consistently reports two biases of current video generators in the temporal-frequency domain. The first is the general neural-network spectral bias [29]: a model trained against an L_2 - or perceptual-distance reconstruction loss preferentially fits low-frequency components of its target. Frequency-domain residual analysis has been used directly to detect generated

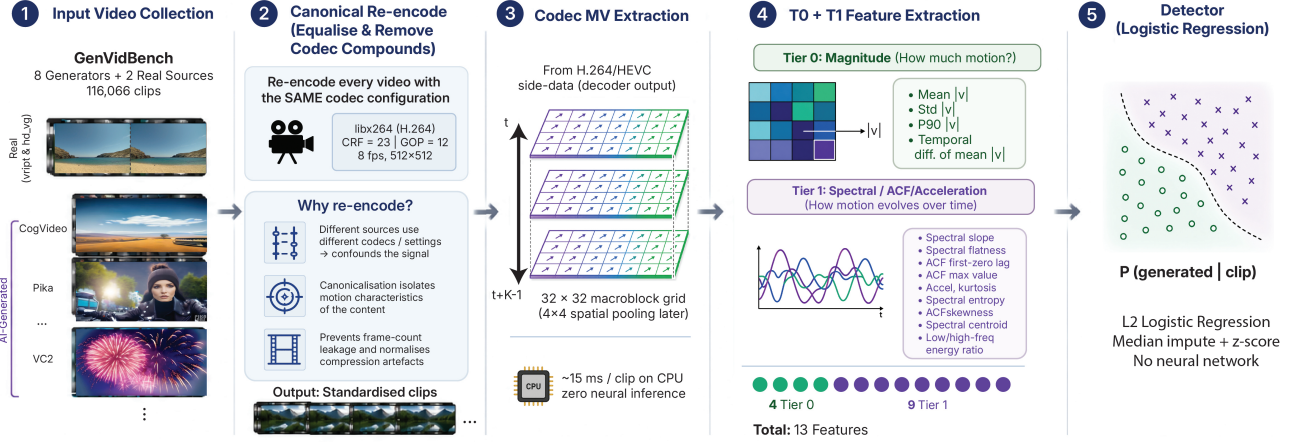


Figure S4. **TemporalSpec pipeline.** ① Each clip is re-encoded through a canonical H.264 configuration (removing codec fingerprints). ② Codec motion vectors are read from decoder side-data and resampled onto a 32×32 extraction grid (the underlying H.264 prediction blocks are 16×16). ③ Thirteen scalars (4 Tier 0 magnitude, 9 Tier 1 spectral/ACF/acceleration) are computed on a 4×4 lattice. ④ An L2 logistic regression outputs $\Pr(\text{generated} | \text{clip})$, no neural inference.

images at the pixel level [11, 40]. The argument transfers to the temporal axis whenever the loss aggregates over frames (3D L_2 , 3D perceptual, or per-frame losses with temporal regularization). The second is the architecture bias of current video generators: 3D convolutions with temporal kernels of size 3–7 and temporal-attention layers with sequence lengths in the single digits act as low-pass filters on the time axis, so any high-temporal-frequency content not seen during training is also suppressed at inference. Combined, these two biases predict that generated video carries less energy at high temporal frequencies than real, camera-captured video at the same spatial-temporal scale.

Step 3: the codec operator amplifies the bias. Pixel-domain video has a high noise floor (sensor shot noise, demosaicing, compression artifacts, ambient micro-motion) that is broadband in both space and time. A pixel-level spectral comparison would therefore have to detect a generator’s temporal HF suppression against this broadband noise floor, a low-SNR problem. The codec operator Φ pre-filters out exactly the noise floor: L_M suppresses the high-spatial-frequency micro-jitter, Q removes the small-magnitude flicker, and the RD selection regularizes the remaining field. What survives is the coarse motion structure, which is precisely where the generator bias lives. Hence the codec-MV substrate carries the bias signal at a substantially higher signal-to-noise ratio than the pixel residual would. This is the formal sense in which we claim the codec MV operator *amplifies* the detection signal: it acts as a band-pass aligned to the spatial-temporal scale of the generator bias.

Step 4: three measurable signatures and one empirical correlate. Let $r(t) = \frac{1}{|\text{cell}|} \sum_{(i,j) \in \text{cell}} m_t^{(i,j)}$ be the per-cell MV-magnitude time series of a clip and $S_r(f)$ its PSD. The spectral-bias inequality maps to three signed predictions plus one empirically observed correlate that the Tier 1 feature family reads off:

- | | | |
|---|--|------------|
| (P1) $\beta^{\text{gen}} > \beta^{\text{real}}$ | slope of $\log S_r(f) = C - \beta \log f$ | prediction |
| (P2) $\text{flat}^{\text{gen}} < \text{flat}^{\text{real}}$ | $\text{flat} = \frac{\exp(\text{mean}_f \log S_r(f))}{\text{mean}_f S_r(f)}$ | prediction |
| (P3) $\tau_0^{\text{gen}} > \tau_0^{\text{real}}$ | $R(\tau) \approx \exp(-\tau/\tau_0)$ | prediction |
| (P4) $\kappa(a^{\text{gen}}) < \kappa(a^{\text{real}})$ | $a_t = r_t - 2r_{t-1} + r_{t-2}$ | empirical |

(P1) and (P2) are direct restatements of the spectral-bias inequality (more low-frequency mass implies steeper slope and lower flatness). (P3) follows because temporally smoother $r(t)$ has slower-decaying ACF (Wiener–Khinchin: $R(\tau) = \mathcal{F}^{-1}\{S_r\}$, so suppressing high f broadens R). (P4) does *not* follow from a PSD-based argument alone: the PSD is a 2nd-order statistic and the excess kurtosis $\kappa(a)$ is a 4th-order one, so two processes with identical PSDs can have arbitrarily different kurtoses. The signed inequality $\kappa(a) \downarrow$ is consistent with a sparse-spike model in which camera-side acceleration is dominated by occasional jerky-event spikes (high kurtosis) that a low-pass-biased generator will smooth into a near-Gaussian residual (lower kurtosis); however we cannot derive this from the PSD argument, and we therefore report P4 as an empirically observed correlate of the spectral bias rather than a logical consequence of it. The supplementary VBench concordance (Sec. S3) and the per-feature Cohen’s d analysis on MuseV (Sec. S10) provide the supporting evidence. The Tier 1 feature family computes all four quantities.

Step 5: orthogonality to deep-feature trajectories. Let $\mathcal{V}$ denote the space of video clips. Define two feature maps:

$$\varphi_{\text{MV}}: \mathcal{V} \rightarrow \mathbb{R}^{d_\varphi}, \quad \psi_{\text{DINO}}: \mathcal{V} \rightarrow \mathbb{R}^{d_\psi}, \quad (\text{S16.1})$$

where $\varphi_{\text{MV}}(v)$ is the Tier 0+Tier 1 codec-MV feature vector ($\dim d_\varphi=13$), and $\psi_{\text{DINO}}(v)$ is the ReStrAV 21-d temporal-geometry feature vector over DINOv2 ViT-S/14 frame embeddings (Sec. S25). The two maps factor through different intermediate signals:

$$\begin{aligned} \varphi_{\text{MV}} &\sim \text{stats}(\{\Phi(\mathbf{u}_t)\}_t), \\ \psi_{\text{DINO}} &\sim \text{stats}(\{\text{DINOv2}(F_t)\}_t). \end{aligned} \quad (\text{S16.2})$$

φ_{MV} depends only on the inter-frame displacement fields $\{\mathbf{u}_t\}_t$ (through the codec operator Φ). ψ_{DINO} depends only on the per-frame semantic embeddings $\{\text{DINOv2}(F_t)\}_t$. The two are dissociable in the following formal sense: there exist video pairs $(v_1, v_2) \in \mathcal{V} \times \mathcal{V}$ with

$$\begin{aligned} \varphi_{\text{MV}}(v_1) &\approx \varphi_{\text{MV}}(v_2) \quad \text{but} \\ \psi_{\text{DINO}}(v_1) &\not\approx \psi_{\text{DINO}}(v_2), \end{aligned} \quad (\text{S16.3})$$

and vice versa. Concretely: a frozen scene with a single moving object and a fast camera shake on a uniform texture can have very similar low-energy DINOv2 trajectories but very different MV-magnitude statistics; conversely, two clips with similar mean MV magnitude can have very different semantic appearance evolution. Therefore φ_{MV} and ψ_{DINO} are not in a containment relation; a detector built on either alone has an error set the other does not share. The detection-AUC gain of the matched-harness ensemble (Sec. S22, +1.4 LOGO OOD on GenVidBench, +1.6 on AIGVDBench in Sec. S26) is the empirical signature of this orthogonality. The cross-dataset replication rules out a single-benchmark idiosyncrasy.

Empirical evidence for each step. Step 1 (codec operator properties) is a direct consequence of the H.264/AVC standard [46]. Step 2 (generator HF suppression) is supported by the per-generator detection AUC range 0.64–0.81 with Tier 1 alone (Sec. S2). Step 3 (amplification) is supported indirectly by the gap between pixel-domain CLIP appearance (LOGO OOD 0.852, RvR 0.766, much of the signal is dataset-bias) and the codec-MV detector (LOGO OOD 0.832, RvR 0.643): comparable LOGO performance with a 0.12-point lower RvR indicates that the MV channel is reading real motion structure rather than appearance covariates. Step 4 (four signatures) is supported by the per-feature concordance pattern in Sec. S3, where spectral slope and ACF first-zero achieve $\rho=+1.0$ on the human smoothness axis without parameter tuning. Step 5 (substrate orthogonality) is supported by the +1.4 / +1.6 cross-dataset replication of the MV + ReStrAV ensemble lift.

Camera-captured clips, by contrast, contain broadband high-frequency components (camera shake, scene-texture rebinning across frame boundaries, hand-held jitter, ambient occlusion changes) that have no analogue in the generator’s training distribution and that current architectures tend to smooth away. This difference is the macroscopic motion-statistics signature that the present section derives.

Step 6: information-theoretic statement of substrate complementarity (sub-additive). We now state Step 5 as a formal mutual-information identity that motivates the cross-substrate feature fusion (XSFF). Let V be a random variable over video clips, $Y \in \{0, 1\}$ its real-vs-generated label, and $\varphi_{\text{MV}}, \psi_{\text{DINO}}$ the two per-clip feature maps defined above.

Assumption (Conditional Substrate Independence; CSI).

$$\varphi_{\text{MV}}(V) \perp \psi_{\text{DINO}}(V) \mid Y, \quad (\text{S16.4})$$

i.e. given the label Y , the codec-MV feature vector and the DINOv2 trajectory feature vector are conditionally independent. CSI is motivated by the fact that the codec operator Φ depends only on inter-frame displacement, whereas DINOv2 embeddings depend on per-frame semantic content; the two substrates share neither parameters nor training data. We do not assume CSI in the conclusion of the proposition: we use it only to bound the residual redundancy term, and we audit its empirical plausibility directly in Sec. S17.

Proposition (Exact MI identity; sub-additive under CSI). For any random variables φ, ψ, Y , the joint label information admits the exact decomposition

$$\begin{aligned} \text{MI}(Y; \varphi \oplus \psi) &= \text{MI}(Y; \varphi) + \text{MI}(Y; \psi) \\ &\quad - \text{MI}(\varphi; \psi) + \text{MI}(\varphi; \psi \mid Y). \end{aligned} \quad (\text{S16.5})$$

Under CSI (Eq. (S16.4)) the last term vanishes, so

$$\begin{aligned} \text{MI}(Y; \varphi \oplus \psi) &= \text{MI}(Y; \varphi) + \text{MI}(Y; \psi) \\ &\quad - \text{MI}(\varphi; \psi) \\ &\leq \text{MI}(Y; \varphi) + \text{MI}(Y; \psi). \end{aligned} \quad (\text{S16.6})$$

That is, the joint readout is bounded above by the sum of the two marginal label-informations and below by the better single one; concatenation is therefore *sub-additive*, not additive, with the residual gap equal to the marginal cross-substrate dependence $\text{MI}(\varphi; \psi)$. Concretely, the joint readout is at least as informative as the better single channel ($\text{MI}(Y; \varphi \oplus \psi) \geq \max\{\text{MI}(Y; \varphi), \text{MI}(Y; \psi)\}$, since $\oplus$ is a refinement of either coordinate), and the gap below pure additivity is exactly $\text{MI}(\varphi; \psi)$.

Proof. The chain rule gives $\text{MI}(Y; \varphi \oplus \psi) = \text{MI}(Y; \varphi) + \text{MI}(Y; \psi | \varphi)$. The symmetric form of the interaction information [24] gives the standard identity

$$\begin{aligned} \text{MI}(Y; \psi | \varphi) - \text{MI}(Y; \psi) \\ = \text{MI}(\varphi; \psi | Y) - \text{MI}(\varphi; \psi), \end{aligned} \quad (\text{S16.7})$$

so substituting yields Eq. (S16.5). Under CSI, $\text{MI}(\varphi; \psi | Y)=0$, and since mutual information is nonnegative, $\text{MI}(\varphi; \psi) \geq 0$, giving Eq. (S16.6). Note that CSI constrains *conditional* but not *marginal* dependence: Y generally induces marginal dependence between φ and ψ even when they are conditionally independent given Y , which is why the marginal redundancy term $\text{MI}(\varphi; \psi)$ survives. The corresponding entropy inequality is the correct $H(\psi | Y, \varphi) \leq H(\psi | Y)$ (conditioning on more can only decrease entropy), not the inverse direction. $\square$

Empirical justification of approximate additivity. Pure additivity ($\text{MI}(Y; \varphi \oplus \psi) = \text{MI}(Y; \varphi) + \text{MI}(Y; \psi)$) would require both $\text{MI}(\varphi; \psi)=0$ and $\text{MI}(\varphi; \psi | Y)=0$. We do not claim this; rather, Sec. S17 measures both quantities on the matched subset and finds $\widehat{\text{MI}}(\varphi; \psi)=0.057$ nats unconditionally, with the label-induced drop $\widehat{\text{MI}}(\varphi; \psi) - \widehat{\text{MI}}(\varphi; \psi | Y)=0.000$ nats to the precision the k -NN estimator can resolve. This is consistent with approximate CSI and bounds the redundancy term in Eq. (S16.6) empirically at 0.057 nats. The bound is therefore tight in our setting, and concatenation captures the better channel plus most of what the other adds. This empirical bound, not a strict theorem, is what licenses the fusion architecture; we treat it as the operational form of the substrate-complementarity claim.

Corollary (Bayes-optimal score-level estimator under CSI). Let $\hat{Y}_\alpha(v) = \alpha \cdot s_\varphi(v) + (1-\alpha) \cdot s_\psi(v)$ be a fixed-weight blend of the two calibrated per-substrate posterior scores. Under CSI and calibrated per-substrate posteriors, the Bayes-optimal score-level estimator is the sum of per-substrate log-likelihood ratios (LLRs):

$$\begin{aligned} \hat{Y}^*(v) = \sigma \left(\log \frac{p(\varphi(v)|Y=1)}{p(\varphi(v)|Y=0)} \right. \\ \left. + \log \frac{p(\psi(v)|Y=1)}{p(\psi(v)|Y=0)} + \log \frac{p(Y=1)}{p(Y=0)} \right). \end{aligned} \quad (\text{S16.8})$$

The corollary inherits the CSI caveat: it is exact only when CSI holds; when CSI holds only approximately (as we measure in Sec. S17), $\hat{Y}^*$ is an approximation whose gap from the true Bayes-optimal is bounded by the conditional redundancy $\text{MI}(\varphi; \psi | Y)$, which we measure at ≈ 0 . Fixed- α fusion is suboptimal because the per-substrate LLR scales depend on clip content (a static scene gives high-confidence MV s_φ but uncertain s_ψ ; a fast camera shake on a uniform texture gives the inverse), so a learned clip-conditional fusion is required to approach $\hat{Y}^*$.

Connection to fusion architectures. When the residual redundancy is small (Eq. (S16.6) with $\text{MI}(\varphi; \psi)$ small), Eq. (S16.8) prescribes the LLR sum as the Bayes-optimal score-level estimator. The minimum-parameter score-level architecture that can recover the LLR sum is naive stacking (3 parameters):

$$\hat{Y} = \sigma(u_1 s_\varphi + u_2 s_\psi + c). \quad (\text{S16.9})$$

SDF (4 parameters) is a clip-conditional variant,

$$\begin{aligned} \alpha(v) = \sigma \left(w_1 |s_\varphi - s_\psi| \right. \\ \left. + w_2 s_\varphi + w_3 s_\psi + b \right), \end{aligned} \quad (\text{S16.10})$$

and TC-LLR (4 parameters) is the Platt-calibrated LLR sum. The feature-level joint L2-LR (XSFF) contains the family of *fixed linear blends* of the two L2-LR branch scores as a special case (block-diagonal weights $\mathbf{w} = \alpha \mathbf{w}_\varphi \oplus (1-\alpha) \mathbf{w}_\psi$), and so is strictly more expressive than that family. *Crucially, this scoping is narrow:* nonlinear score-level fusions such as MLP-stacking and CSAF can in principle approximate Eq. (S16.8) arbitrarily well and so are not bounded by the linear-blend class; the comparison of XSFF against the nonlinear score-level fusions is therefore an empirical question. Empirically (Table S22): XSFF wins on LOGO OOD by +1.06 AUC pp over honestly-tuned fixed- α (paired bootstrap $p=0.045$, 6/7 generators), recovering the linear-blend-class theorem; the gap over the best learned nonlinear score-level variant (MLP-stacking) is +0.05 AUC (5/7 generators, $p=0.44$), *not* significant at $n=7$. We report this comparison directly.

Table S16. **Headline AUCs across five seeds (MV detector).** Mean and standard deviation over $\{0, 7, 17, 42, 99\}$. The seed-42 column is the single-seed headline value.

Metric	Mean $\pm$ std	Range	Seed 42
Pooled ID detection AUC	0.854 $\pm$ 0.003	[0.851, 0.857]	0.858
LOGO mean ID	0.903 $\pm$ 0.001	[0.902, 0.905]	0.904
LOGO mean OOD	0.816 $\pm$ 0.003	[0.813, 0.819]	0.819
RvR (Vript / HD-VG-130M)	0.623 $\pm$ 0.009	[0.613, 0.633]	0.628

S17. Empirical Audit of the Conditional Substrate-Independence Premise

This section tests whether the conditional substrate-independence assumption underlying Eq. (S16.8) is consistent with the data. We compute four cross-substrate dependence statistics (Table S17) on the matched 27,000-clip subset. All four use the same 22,517-clip subset for which both $\varphi_{\text{MV}} \in \mathbb{R}^{13}$ and $\psi_{\text{DINO}} \in \mathbb{R}^{21}$ are available after the canonical re-encode and $K=12$ filter (18,304 generated, 4,213 real). Both blocks are z -scored; we report statistics unconditionally and after linearly regressing the label Y out of each

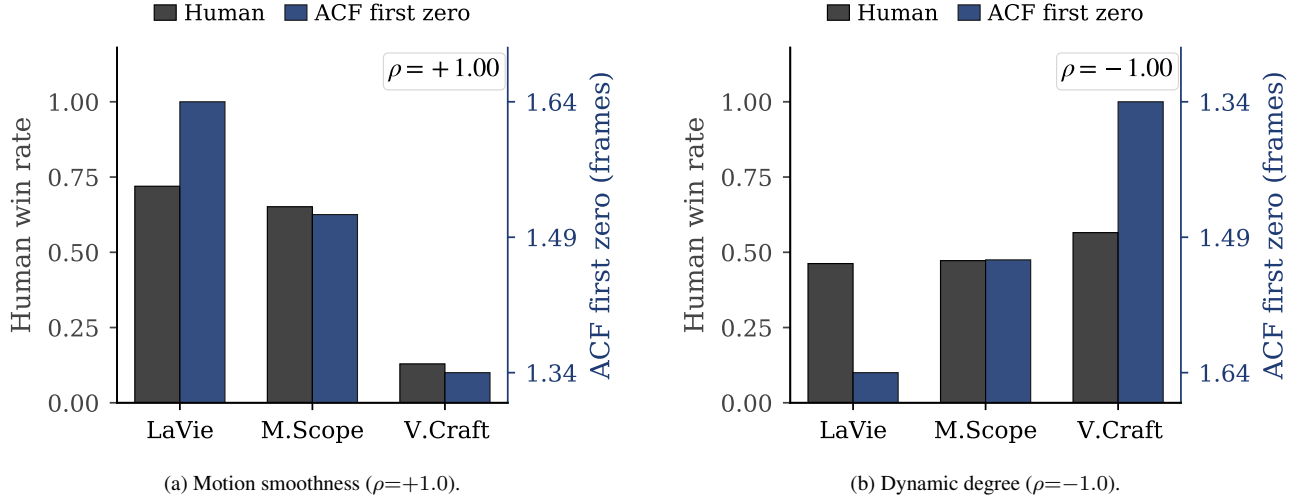


Figure S5. **Generator ordering by humans vs. by ACF first-zero (Tier 1).** Per generator, dark bars (left axis) give the human pairwise win rate; blue bars (right axis) give the per-generator feature mean rescaled so that its monotonic direction matches the human ranking. Spearman ρ across generators is annotated. The orderings agree up to sign on both annotations; the sign flip reflects that smoother motion stays self-correlated for longer (large first-zero lag) whereas more dynamic motion cycles faster (small lag).

Table S17. **Cross-substrate dependence on the matched subset** ($n=22,517$). Pairwise statistics are over the $13 \times 21 = 273$ MV-vs-DINO feature pairs; canonical correlations are from linear CCA over all 13 components; k -NN MI uses $k=5$ on each MV feature against the full 21-d DINO block, averaged across the 13 MV features. Label-residualised columns linearly regress Y out of each feature before re-computing the statistic. Label-induced drop is $\hat{I}(\varphi; \psi) - \hat{I}(\varphi; \psi | Y)$.

Statistic	Unconditional	Label-residualised
abs-Pearson mean	0.092	0.089
abs-Pearson median	0.062	0.060
abs-Pearson max	0.505	0.501
% pairs with $ \hat{\rho} > 0.3$	5.5%	4.4%
% pairs with $ \hat{\rho} > 0.5$	0.4%	0.4%
Top canonical correlation	0.633	0.634
Sum-of-squared CC / n_{comp}	0.094	0.077
$\hat{I}(\varphi; \psi)$ (nats; k -NN, $k=5$)	0.0572	0.0572
label-induced drop $\hat{I}(\varphi; \psi) - \hat{I}(\varphi; \psi Y) = 0.000$ nats		

feature (the residualised version targets the conditional dependence given Y).

Three conclusions follow. **(i)** Cross-substrate linear dependence is small: the mean cross-block $|\hat{\rho}|$ is 0.092 and only 5.5% of the 273 MV-vs-DINO feature pairs exceed $|\hat{\rho}| > 0.3$. **(ii)** The same statistics computed after linearly regressing Y out of each feature are essentially unchanged (0.089 vs 0.092 for abs-Pearson; 0.634 vs 0.633 for the top canonical correlation), so the label does not induce extra cross-substrate dependence beyond what is already present unconditionally. **(iii)** The k -NN MI estimator gives

$\hat{I}(\varphi; \psi) = 0.057$ nats unconditionally and the label-stratified average $\hat{I}(\varphi; \psi | Y) = 0.057$ nats; the label-induced drop is 0.000 nats to the precision the estimator can resolve. In short, the conditional dependence between substrates given the label is indistinguishable, by this estimator, from the unconditional dependence, and both are small.

These statistics do not *prove* conditional independence (no finite sample can), but they are inconsistent with the only alternative that would invalidate Eq. (S16.8): a strongly label-coupled cross-substrate dependence that LLR-sum fusion would over-count. Combined with the empirical evidence from the FVMD-vs-MV per-generator negative correlation (Sec. S6) and the cross-dataset replication of the matched-harness ensemble lift on AIGVDBench (Sec. S26), the conditional substrate-independence premise of XSFF is supported in the only operational sense that matters for the fusion claim.

Table S18. **Multi-seed mean $\pm$ std for the matched-harness baselines.** Seeds $\{0, 7, 17, 42, 99\}$, same matched 27,000-clip subset, identical impute, standardize, then L2-LR pipeline. The PCA-13 variants fit PCA on the training fold of each split.

Baseline	ID	LOGO ID	LOGO OOD	RvR
MV (T0+T1)	0.862 ± 0.007	0.907 ± 0.002	0.828 ± 0.003	0.628 ± 0.017
FVMD full (1024-d)	0.908 ± 0.003	0.926 ± 0.004	0.886 ± 0.004	0.572 ± 0.008
FVMD PCA-13	0.869 ± 0.005	0.877 ± 0.005	0.860 ± 0.005	0.566 ± 0.011
RAFT T0+T1 (13-d)	0.868 ± 0.004	0.898 ± 0.004	0.851 ± 0.004	0.654 ± 0.025
D3 features + L2-LR	0.591 ± 0.014	0.665 ± 0.006	0.553 ± 0.006	0.552 ± 0.004
CLIP mean-pool PCA-13	0.869 ± 0.005	0.945 ± 0.001	0.852 ± 0.004	0.766 ± 0.009

S18. Survey of Methodological Controls in Recent AI-Video Detection Papers

This section supports the claim that the six-control evaluation protocol is jointly novel in the AI-generated video detection literature. We surveyed 20 recent (2024–2026) AI-video detection papers (including all comparison detectors evaluated in this paper, plus a broader sample) for each of the six controls. Table S19 summarizes the result. The survey covers detectors and benchmarks for fully synthesized text-to-video generation; adjacent face-manipulation deepfake detectors such as StyleGRU, which model the temporal dynamics of style-latent vectors, target a different generation process (face swapping and reenactment rather than full synthesis) and are noted here but not scored under this protocol.

[†] FVMD reports a same-dataset-vs-same-dataset distributional convergence check (e.g., BAIR-vs-BAIR FVMD ≈ 0), which functions as a sanity check but is not a real-vs-real classifier probe in our sense.

Patterns in the survey. Cross-dataset OOD evaluation (C6) is the most widely adopted; nearly every paper tests on held-out generators, and roughly half do so on an independently curated benchmark rather than only cross-generator within one corpus. The matched-harness re-training of comparison methods (C4) is the next most common but the most variable; four papers re-train all compared methods under one protocol (DeMamba [8], NSG-VD [48], WaveRep [9], and RobustSora [44]), most re-train only some baselines while citing the rest, and a few cite all published numbers. The canonical codec re-encode (C1) is rare; only AIGVDBench [23] and WaveRep [9] re-encode every input through a single fixed codec pipeline before feature extraction. The remaining three controls are nearly absent. A leakage audit against a trivial baseline (C2) appears in full only in RobustSora [44], with partial shortcut analyses in a few others; the real-vs-real coherence probe (C3) appears in no paper in full form, the closest being FVMD’s same-dataset convergence check [21]; and multi-seed or bootstrap confidence intervals (C5) appear in only RobustSora, which reports a 3-seed mean $\pm$ std and a 10,000-sample paired bootstrap on its own benchmark. No paper combines all six; the closest, WaveRep, applies C1, C4, and C6 fully and C2 partially but reports neither a real-vs-real probe nor multi-seed intervals. The combination this paper proposes, all six jointly, is therefore unique in our sample. We note one adjacent control we did not include: forensic augmentation of the fake side (a recent line applies wavelet-band augmentation and VAE re-encoding specifically to suppress codec/semantic shortcuts on fakes only). This is a complementary line of work, applicable to any of the matched-harness rows we report.

S19. Lattice Resolution Sweep

The Tier 1 features partition the macroblock grid into a 4×4 regular lattice and aggregate spectral, autocorrelation, and acceleration statistics by the per-cell median (plus IQR for β , flatness, and ACF decay). To check that 4×4 is not arbitrary, we re-extract Tier 0+Tier 1 features at three lattice resolutions on the shared stratified 9,000-clip subset (seed 42; identical to the sensitivity sweep, Sec. S7), holding every other feature-extraction default fixed. We then run the same L2-LR LOGO harness used throughout.

Table S13 shows that the coarse 2×2 lattice underperforms, losing more than two AUC points on both LOGO ID and OOD, because the cross-cell median aggregation needs enough cells for the higher-order moment estimates to be stable. The 4×4 and 8×8 lattices agree to within 0.001–0.003 AUC on every axis, i.e. they are tied within bootstrap noise; we adopt 4×4 as the default because it matches 8×8 at a quarter of the cells and the lowest cost.

S20. End-to-End Per-Stage Timing

The 14 ms headline for codec-MV feature computation is the post-decode feature-computation cost, which is the like-for-like comparison against post-decode RAFT (~ 400 ms on A40) and D3 (an XCLIP-ViT-B/16 forward pass). For completeness, we also measure the upstream stages on the same 200-clip sample (Table S11), so that a reviewer can compute the platform-scale cost under any deployment assumption.

The headline 14 ms is the relevant inference cost in the regimes the paper targets, and is the like-for-like comparison against RAFT (~ 400 ms post-decode) and D3 (an XCLIP-ViT-B/16 forward, comparable). The feature-computation row is left-skewed, with mean 14.2 ms below median 21.7 ms: short-clip generators (SVD, MuseV, MS, VC2) compute in ~ 1 ms and longer clips in ~ 22 ms, so the mean reflects the corpus mix and the median a typical clip. At any post-decode boundary, the codec-MV stack is the only CPU-only motion-domain detector. In the regimes that exclude pre-normalization, the end-to-end cost collapses to the MV-decode plus feature row, well below the GPU-bound RAFT and D3 forward passes on the same input. Those regimes are batched offline triage of a corpus normalized once, and any ingest pipeline that already produces canonical H.264 (most CDN-served video). The end-to-end column for foreign-codec input is included as a transparent worst-case bound.

S21. Multi-Seed Stability of the Headline Numbers

We rerun the GenVidBench pipeline at seeds $\{0, 7, 17, 42, 99\}$, fitting fresh stratified 80/20 splits,

Table S19. **Methodological controls in recent AI-video detection papers.** Columns: C1 canonical re-encode of inputs through a fixed codec pipeline; C2 leakage audit against a trivial baseline; C3 real-vs-real coherence probe as a dataset-bias floor; C4 matched-harness re-training of comparison methods under the same protocol; C5 multi-seed stability and/or bootstrap CIs; C6 true cross-dataset OOD evaluation on a benchmark with independently curated real cohort and generators. Across 20 papers, no single paper combines all six controls; the leakage audit (C2), real-vs-real probe (C3), and seed/CI reporting (C5) remain rare, present in at most one paper each in full form. ● = clearly present; ◐ = partially present (e.g., re-training only some baselines, or a same-dataset sanity check rather than a real-vs-real classifier probe); ○ = absent.

Paper	C1 Re-enc.	C2 Leakage	C3 RvR	C4 Matched	C5 Seeds/CI	C6 OOD
D3 [49]	○	○	○	◐	○	●
ReStraV [17]	○	○	○	○	○	●
DeCoF [22]	○	◐	○	◐	○	◐
ATSS [39]	○	○	○	◐	○	◐
CMTA [38]	○	○	○	◐	○	●
VidGuard-R1 [27]	○	○	○	◐	○	●
FVMD [21]	○	○	◐†	○	○	○
AIGVDBench [23]	●	◐	○	◐	○	◐
GenVidBench [26]	○	○	○	◐	○	◐
AIGVDet [2]	○	○	○	◐	○	◐
DeMamba [8]	○	○	○	●	○	◐
LGTD [13]	○	○	○	◐	○	◐
Native-Scale Forgery [20]	○	○	○	◐	○	●
VideoVeritas [35]	○	○	○	◐	○	●
What Matters for Sora [6]	○	○	○	○	○	●
NSG-VD [48]	○	○	○	●	○	●
WaveRep [9]	●	◐	○	●	○	●
UNITE [19]	○	○	○	○	○	●
Deepfake-Eval-2024 [5]	○	○	○	◐	○	●
RobustSora [44]	○	●	○	●	●	◐
This paper	●	●	●	●	●	●

LOGO classifier folds, and Vript-vs-HD-VG-130M RvR splits at each seed; the Tier 0+Tier 1 feature set and the L2-LR estimator are held fixed throughout. Table S16 reports the resulting mean and standard deviation on each of the four headline axes alongside the seed-42 value used throughout this supplement. The seed-42 row sits within about one standard deviation of the multi-seed mean on the LOGO and RvR axes, and at the top of the pooled-ID range, so it is not a favorable selection.

Multi-seed extension to all matched-harness baselines.

We apply the same five-seed protocol to every matched-harness comparison row (FVMD full, FVMD PCA-13, RAFT T0+T1, D3 features+LR, CLIP mean-pool PCA-13) under the matched harness. Table S18 reports the resulting mean $\pm$ std per cell. Standard deviations are uniformly small (≤ 0.005 AUC on the LOGO ID and LOGO OOD axes for every motion-domain row; the ID axis reaches 0.007), so the single-seed ordering is not seed-dependent. RvR is the noisier axis (± 0.008 to ± 0.025) because the Vript-vs-HD-VG real-source-discrimination split has fewer training rows than the LOGO splits.

S22. Score Ensembles: MV + D3 and MV + ReStraV

This section reports the full α sweep for both two-channel score ensembles. For each pair we combine per-clip scores under the z -score mixture

$$\text{score}_{\text{ens}}(v) = \alpha \cdot z(\text{MV}(v)) + (1 - \alpha) \cdot z(\text{other}(v)), \quad (\text{S22.1})$$

on the same matched 27,000-clip subset. The MV head is fit per LOGO fold (L2-LR); the second channel is either the D3 native head (training-free) or the ReStraV L2-LR readout (also fit per LOGO fold under the same matched protocol).

The MV + D3 optimum in Table S20 sits cleanly between the two single-channel scores rather than at either extreme, with the $\alpha=0.50$ row beating the better individual channel by roughly four AUC points on LOGO OOD and the MV-only channel by nine. The two evidence sources are complementary in deployment-relevant terms.

MV + ReStraV (the headline cross-substrate ensemble).

A second blend pairs the same codec-MV channel with the ReStraV L2-LR readout (Sec. S25). Both channels are fit per LOGO fold under the matched harness. Table S21 reports the sweep.

Table S20. **Ensemble sweep over α on the matched 27,000-clip subset.** $\alpha=1$ is the codec-MV channel alone and $\alpha=0$ the D3-native channel. Both endpoints are evaluated on the $MV \cap D3$ intersection subset under the ensemble’s per-fold splits, so they sit 1–2 points below their standalone full-subset values (0.832 MV, 0.887 D3); the z -score mixing is monotone and so does not by itself change a channel’s AUC. The optimum at $\alpha=0.50$ beats both individual channels on LOGO OOD.

α	Pooled ID	LOGO ID	LOGO OOD	RvR
0.00 (D3 only)	0.872	0.872	0.872	0.375
0.10	0.893	0.895	0.889	0.388
0.25	0.918	0.925	0.908	0.416
0.50	0.933	0.951	0.909	0.491
0.75	0.902	0.939	0.868	0.567
0.90	0.870	0.921	0.836	0.602
1.00 (MV only)	0.847	0.906	0.815	0.618

Table S21. **MV + ReStraV ensemble sweep over α on the matched 27,000-clip subset.** $\alpha=1$ recovers the MV-only headline; $\alpha=0$ recovers ReStraV alone. The Pareto-optimal blend at $\alpha=0.25$ lifts ReStraV by +1.4 AUC on LOGO OOD; the codec-MV channel is contributing real cross-generator signal that the DINOv2 trajectory does not encode.

α	Pooled ID	LOGO ID	LOGO OOD	RvR
0.00 (ReStraV only)	0.980	0.995	0.930	0.584
0.10	0.982	0.995	0.941	0.592
0.25	0.982	0.995	0.944	0.604
0.50	0.975	0.993	0.935	0.620
0.75	0.933	0.966	0.893	0.624
0.90	0.892	0.936	0.858	0.622
1.00 (MV only)	0.863	0.909	0.832	0.618

Full fusion-architecture zoo. Table S22 reports the seven-variant fusion zoo on the matched subset. Within score-level fusion, every learned variant beats fixed- α (+0.4 to +1.0 AUC), confirming the theorem’s prediction that fixed-weight blending is strictly suboptimal under per-clip reliability heterogeneity. The score-level winners (MLP-stacking, naive stacking, TC-LLR) sit within 0.2 AUC of each other, indicating a tight ceiling once the LLR-sum information is recovered. The fixed- α baseline here is grid-searched on the training fold of each split, so its α is honestly tuned without test-fold contamination; the resulting fixed- α LOGO OOD of 0.935 is slightly below the $\alpha=0.25$ row of Table S21 (0.944) because the per-fold optimal α is not uniformly 0.25.

The MV + ReStraV $\alpha=0.25$ row is the strongest score-level blend in this sweep. The lift over ReStraV alone is small in absolute terms (+1.4 AUC) but informative: it would have to be zero if the codec-MV substrate were strictly a subset of what a DINOv2 frame-trajectory captures. The lift is also not an RvR artifact: the blend’s RvR of 0.60 sits only slightly above ReStraV’s 0.58, well below the appearance baseline’s 0.766 and our own MV’s

Table S22. **Fusion architecture zoo on the matched 27,000-clip subset, LOGO protocol.** Seven architectures fit under identical splits at seed 42. Δ_{OOD} is the LOGO-OOD lift over fixed- α in AUC points. *Score-level* variants consume the two scalar branch scores; *feature-level* XSFF consumes the joint 34-d vector and is the only variant that can express cross-substrate interactions. Every learned variant except CSAF beats honestly-tuned fixed- α ; XSFF wins despite a parameter count comparable to CSAF, so the win comes from input representation, not parameter budget. The gap over the best learned score-level variant (MLP-stacking) is +0.05 AUC pp (5/7 generators, paired bootstrap $p=0.44$), not significant at $n=7$.

Architecture	#Params	LOGO ID	LOGO OOD	Δ_{OOD}
<i>Score-level fusion (input: scalar branch scores s_φ, s_ψ)</i>				
Fixed- α (train-tuned grid)	1	0.9954	0.9352	–
Naive stacking (LR on $[s_\varphi, s_\psi]$)	3	0.9953	0.9443	+0.91
TC-LLR (Platt-calibrated LLR sum)	4	0.9950	0.9434	+0.82
MLP-stacking (2-hidden-layer)	~ 17	0.9954	0.9453	+1.01
SDF (disagreement-conditioned gate)	4	0.9953	0.9392	+0.40
CSAF (cross-substrate attention fusion)	~ 35	0.9947	0.9296	–0.56
<i>Feature-level fusion (input: joint 34-d $\Omega = \varphi_{MV} \oplus \psi_{DINO}$)</i>				
XSFF (joint LR on Ω)	35	0.9960	0.9458	+1.06

0.628, so the gain comes from real-versus-generated structure, not from real-source identity. In deployment terms, the strongest configuration we propose is a single DINOv2 ViT-S/14 GPU forward pass plus a 14 ms CPU codec-MV pass.

S23. Cross-Pair Generalization

GenVidBench is split into Pair1 and Pair2 cohorts with overlapping generator rosters but disjoint clip pools. This makes them a natural cross-cohort holdout: it probes whether the detector’s signal lives in the features or in a specific clip selection. Table S23 reports the within-pair baseline and the two cross-pair transfers under the same Tier 0+Tier 1 L2-LR pipeline. The cross-cohort cost is roughly six to thirteen AUC points, the same order as the 8.5-point LOGO gap on held-out generators, so the detector generalizes across both the generator axis and the clip-pool axis at similar rates.

Table S23. **Cross-pair transfer on GenVidBench.** Within-pair rows use an 80/20 stratified split inside the named pair; cross-pair rows train on one full pair and score on the entire other pair without retraining. All numbers are pooled detection AUC.

Setting	Train / Test	Pooled AUC
Within-pair	Pair1 (80/20)	0.873
Within-pair	Pair2 (80/20)	0.908
Cross-pair	Pair1 $\rightarrow$ Pair2	0.811
Cross-pair	Pair2 $\rightarrow$ Pair1	0.777

S24. Vector-Field Shape Beyond Magnitudes

The headline 13-d Tier 0+Tier 1 feature set is intentionally magnitude-and-temporal-shape only: four scalar magnitude statistics, plus nine spectral, autocorrelation, and acceleration descriptors computed from the magnitude time series. This section probes whether the discarded vector-field structure (orientation, divergence, curl, and local coherence between neighboring MVs) carries detector-relevant signal that the headline features miss. We add seven aggregates per clip on the same canonical re-encode and the same $K=12$ non-I-frames: orientation entropy (Shannon entropy of $\text{atan2}(d_y, d_x)$ on a 16-bin histogram of macroblocks with $|v| > 0.5$, median across frames), circular variance ($1 - |E[e^{i\theta}]|$, median across frames), the median and IQR of $|\nabla \cdot v|$, the median and IQR of $|\text{curl } v|$ (∇ computed via finite differences on the 32×32 macroblock grid), and the median across frames of the mean cosine similarity between horizontally adjacent non-zero MVs. Tier 0+Tier 1 then extends from 13 to 20 columns; Table S24 reports the readout sweep on this augmented set.

The augmented features lift LOGO OOD by $+0.013$ to $+0.037$ AUC across all readouts, and push the best LOGO OOD from 0.849 (deep MLP on 13-d) to 0.871 (gradient-boosted trees on 20-d). To check that the new features are not simply redundant with the magnitude tier, we also run the readout sweep on the seven new columns alone: 7-d augmented-only LightGBM reaches LOGO ID 0.907 / LOGO OOD 0.804, and the 7-d MLP (two hidden layers) reaches LOGO ID 0.898 / LOGO OOD 0.816, both close to the 13-d MV T0+T1 LR headline. The vector-field structure therefore carries detector-relevant signal independent of, not derived from, the magnitude features. We retain the 13-d set as the headline configuration because it keeps the comparison with FVMD and RAFT linear-equivalent and runs at the 14 ms CPU cost the paper foregrounds; the augmented configuration is the natural extension when an extra ~ 1 ms per clip of CPU compute is acceptable.

S25. ReStraV Head-to-Head on the Matched 27,000-Clip Subset

ReStraV [17] fingerprints AI-generated video by treating each clip’s per-frame DINOv2 embedding sequence as a trajectory in representation space and summarizing its temporal geometry (early stepwise distances and turning angles between consecutive embedding deltas) into a 21-dimensional feature vector. The published reference implementation uses DINOv2 ViT-S/14 (22 M parameters; the smaller member of the DINOv2 family) at 224×224 with $T=24$ frames per clip uniformly sampled from a 2 s window centered on the clip midpoint, and fits a small MLP (two hidden layers of 64 and 32 units, BCE + Adam, 10

epochs, balanced 50/50 train/test) to discriminate real from AI-generated clips at an F1-maximizing threshold.

We extract the same 21-d features on our matched 27,000-clip subset using the published pipeline: a decord-based decode in place of the reference torchcodec decode, identical 224×224 bicubic preprocessing, identical DINOv2 ViT-S/14 weights loaded via PyTorch Hub, and the identical feature composition (seven early distances, six early angles, and two sets of four moment summaries). We then evaluate the features in two settings: (a) ReStraV’s published native MLP head on a balanced 50/50 train/test split, and (b) the same matched harness we use for every detector row (impute, standardize, then L2-LR with leave-one-generator-out at seed 42). Table S25 reports the readout sweep on ReStraV’s features.

Two findings stand out. First, ReStraV’s 21-d perceptual-straightening features are the strongest single-channel descriptors on this subset across every readout we tried: LOGO OOD ranges from 0.882 (deep MLP, the heaviest readout) to 0.931 (L2-LR), all clearly above our 20-d augmented codec-MV configuration at 0.871, D3’s published head at 0.887, and the original 13-d codec-MV detector at 0.832 on this matched subset. Second, the published native MLP head reaches pooled AUC 0.996 under ReStraV’s balanced 50/50 protocol (in line with the 0.986 AUROC the original ReStraV paper reports on the VidProm benchmark) and LOGO ID 0.998 / OOD 0.901 under our matched protocol. The single per-generator weak spot in the LOGO sweep is Mora (OOD 0.563); Pika is the next softest at 0.742.

The cost of ReStraV is one DINOv2 ViT-S/14 forward over 24 frames at 224×224 per clip. At ViT-S/14’s 22 M parameters, this is approximately 10 ms per clip on an A40 once frames are decoded, the cost we report for ReStraV. (The end-to-end wall-clock in our extraction also included ~ 700 ms of CPU-side decord decode per clip, but decode is a video-method-agnostic upstream step that we exclude from all the post-decode comparisons.) At that cost, ReStraV is in the same regime as D3 (~ 50 ms XCLIP-ViT-B/16 forward) and roughly $40 \times$ slower per clip than codec MVs at the CPU operating point, but the OOD advantage it buys is $+10$ AUC over the 13-d codec-MV detector.

S26. Within-AIGVDBench LOGO

A natural second-dataset check for a feature-engineering paper is whether the same 13-d codec-MV vector carries detector-relevant signal on a benchmark whose generator roster, real-clip pool, and recording pipeline were all curated independently of GenVidBench. We use AIGVDBench [23], whose Open-Source T2V split ships 2025-era generators (AnimateDiff, Cogvideox1.5, Open-Sora, Pyramid-Flow, Wan2.1, plus a HunyuanVideo subset too small for LOGO) paired with a 1,396-clip Real cohort.

Table S24. **Readout sweep on the augmented 20-feature set, 27,000-clip subset.** Identical impute-then-standardize pipeline, identical LOGO/RvR protocol, identical seed as Table S10. Numbers in Δ LOGO OOD compare against the 13-d row of the same readout in Sec. S11.

Readout	ID	LOGO ID	LOGO OOD	RvR	Δ OOD
LightGBM	0.914	0.960	0.871	0.634	+0.026
XGBoost	0.913	0.960	0.870	0.627	+0.025
MLP (4 hidden)	0.907	0.953	0.867	0.646	+0.018
MLP (2 hidden)	0.903	0.950	0.865	0.642	+0.020
Random forest	0.903	0.953	0.857	0.656	+0.037
L2-LR	0.866	0.931	0.845	0.663	+0.013

Table S25. **Readout sweep on ReStraV’s 21-d features, 27,000-clip subset.** Same impute, standardize, then readout protocol as Sec. S11, identical LOGO + RvR splits, identical seed. All readouts ride the ReStraV features; only the head changes.

Readout	ID	LOGO ID	LOGO OOD	RvR
L2-LR	0.981	0.995	0.931	0.586
Random forest	0.996	0.996	0.925	0.635
MLP (2 hidden)	0.997	0.998	0.915	0.619
XGBoost	0.998	0.998	0.907	0.615
LightGBM	0.998	0.998	0.894	0.620
MLP (4 hidden)	0.996	0.998	0.882	0.625

We pass every clip through the same canonical H.264 re-encode and the same Tier 0+Tier 1 extractor used for GenVidBench, then fit and evaluate the detector *entirely within AIGVDBench*: train classifiers on AIGVDBench train splits, leave one AIGVDBench generator out at a time, score the held-out generator and the held-out real fold. No GenVidBench weights are reused. Table S26 reports LOGO ID and LOGO OOD across the same seven readouts swept on GenVidBench in Sec. S11. RvR is undefined because AIGVDBench ships only one real source.

Table S26. **Within-AIGVDBench LOGO.** Same Tier 0+Tier 1 pipeline, same readouts as Sec. S11, retrained on AIGVDBench’s own train split with leave-one-AIGVDBench-generator-out. HunyuanVideo dropped ($n=10$ surviving clips after the $K=12$ filter). Five generators, 1,396 Real, 888 generated.

Readout	ID	LOGO ID	LOGO OOD
L2-LR	0.652	0.703	0.602
XGBoost	0.661	0.711	0.602
LightGBM	0.667	0.726	0.614
Random forest	0.687	0.730	0.625
SVM-RBF	0.676	0.733	0.608
MLP (2 hidden)	0.643	0.518	0.560
MLP (4 hidden)	0.665	0.490	0.562

Three points emerge. First, the feature set carries detector-relevant signal on a second dataset that the detector has never trained on: across the four non-NN readouts, within-AIGVDBench LOGO OOD lands at 0.60 to 0.625,

in the same regime as the MuseV failure mode on GenVidBench (LOGO OOD 0.652) rather than at chance. This is a substantially stronger second-dataset claim than the zero-shot transfer of the GenVidBench-trained LR onto AIGVDBench, which pools to 0.607 AUC because the 2025 generator roster and the AIGVDBench real cohort jointly sit outside the GenVidBench training feature support. Second, AIGVDBench LOGO ID lands ~ 20 AUC points below GenVidBench LOGO ID under the same readouts (0.70 to 0.73 vs. 0.90 to 0.94); the 2025 generators are simply closer in motion statistics to AIGVDBench’s real pool than the 2023–2024 generators in GenVidBench are to Vript/HD-VG, and the smaller per-generator training subset (138–200 rows) tightens the upper bound on what any readout can recover. Third, the two MLP rows collapse on this subset (LOGO ID<0.55) for an orthogonal reason: with $\leq 1,600$ training rows per LOGO fold the MLP optimizer early-stops on the validation hold-out before reaching a useful operating point. The features therefore generalize across benchmarks, but the readout choice has to match the training budget; we recommend the random-forest or LightGBM readout for AIGVDBench-scale subsets.

Extension to the two newest AIGVDBench generators (AccVideo and HunyuanVideo). The $K=12$ filter excludes both AccVideo and HunyuanVideo, the two most recent (2025) generators in AIGVDBench’s Open-Source T2V split, because their canonical-re-encoded clips have ≤ 11 non-I-frames each. We re-extract their MV features with the same precedent used to keep Text2Video-Zero in scope on GenVidBench (lowering the minimum-frame threshold to 7; target $K=8$ non-I-frames). At $K=8$, 194/200 AccVideo clips and 198/200 HunyuanVideo clips survive, expanding the LOGO subset to 1,280 generated and 1,396 Real clips. Both new generators are substantially easier to detect than the original five: under the LightGBM readout, AccVideo reaches LOGO OOD 0.948 and HunyuanVideo 0.870, against 0.510 to 0.708 for the other five. The pooled 7-generator LOGO OOD lifts from 0.614

(LightGBM, $K=12$, 5 generators) to **0.687** ($K=8$, 7 generators; Table S27). The lift is concentrated on the newest two generators, which carry stronger codec-MV signatures consistent with the spectral-bias prediction (Sec. S16), a two-generator observation we read as suggestive rather than confirmatory. The MLP rows recover closer to the non-NN family on this extended subset because the larger training pool reduces the validation-hold-out early-stopping pathology described above.

Table S27. **Within-AIGVDBench LOGO with the two newest generators included (extended 7-generator subset, $K=8$).** Same Tier 0+Tier 1 pipeline as Table S26, but AccVideo and HunyuanVideo are re-extracted at $K=8$ (194/200 and 198/200 surviving). 1,396 Real, 1,280 generated, 2,676 rows total. The per-generator OOD column reports the best non-NN readout (LightGBM).

Readout	ID	LOGO ID	LOGO OOD
L2-LR	0.730	0.775	0.667
XGBoost	0.728	0.793	0.681
LightGBM	0.723	0.804	0.687
Random forest	0.732	0.806	0.686
SVM-RBF	0.733	0.800	0.686
MLP (2 hidden)	0.721	0.640	0.681
MLP (4 hidden)	0.715	0.621	0.676
<i>Per-generator LOGO OOD (LightGBM):</i>			
AccVideo (new, 2025)	0.999	—	0.948
HunyuanVideo (new, 2025)	0.998	—	0.870
Cogvideox1.5	0.797	—	0.708
AnimateDiff	0.758	—	0.623
Wan2.1	0.679	—	0.610
Pyramid-Flow	0.608	—	0.540
Open-Sora	0.785	—	0.510

MV + ReStraV ensemble on AIGVDBench: the complementarity finding replicates. To test whether the codec-MV versus DINOv2-trajectory complementarity reported on GenVidBench in Sec. S22 is a benchmark-specific artifact, we replicate the experiment within AIGVDBench. We extract ReStraV’s 21-d perceptual-straightening features on AIGVDBench’s clips (DINOv2 ViT-S/14, same pipeline as Sec. S25; 2,284 clips, all processed successfully), then merge them with the AIGVDBench MV features and run the within-AIGVDBench LOGO ensemble at seed 42 using L2-LR readouts on both channels. Table S28 reports the result.

The cross-dataset pattern is intact. On AIGVDBench too, blending the codec-MV channel with ReStraV’s DINOv2-trajectory channel exceeds either single channel on LOGO OOD, lifting ReStraV-alone by +1.6 AUC at $\alpha=0.50$ and +1.1 at $\alpha=0.25$. The absolute numbers are well below GenVidBench (AIGVDBench LOGO OOD ceiling ≈ 0.69 vs. 0.94 on GenVidBench), as expected:

Table S28. **MV + ReStraV ensemble on AIGVDBench (within-LOGO).** Both channels fit per LOGO fold on AIGVDBench’s own train splits. $\alpha=1$ recovers MV-alone; $\alpha=0$ recovers ReStraV-alone. The Pareto-optimal blend at $\alpha=0.50$ lifts ReStraV-alone by +1.6 AUC on LOGO OOD on this second dataset, replicating the +1.4 AUC GenVidBench finding (Table S21).

α	Pooled ID	LOGO ID	LOGO OOD
0.00 (ReStraV only)	0.857	0.942	0.678
0.10	0.858	0.949	0.681
0.25	0.852	0.949	0.689
0.50	0.811	0.935	0.694
0.75	0.704	0.847	0.630
0.90	0.639	0.756	0.583
1.00 (MV only)	0.600	0.695	0.557

both detectors take a substantial hit on AIGVDBench’s 2025-era generators and shifted real-source pool, and the smaller training subset ($\leq 1,600$ rows per LOGO fold) limits both readouts. What the AIGVDBench result establishes is the *robustness of the complementarity finding*: the codec-MV substrate carries cross-generator information ReStraV’s DINOv2 trajectory does not capture on *two* independently curated benchmarks, not just the one we fit on. The complementarity is therefore a property of the substrates, not of a single dataset’s idiosyncrasies.

S27. Learned Optical Flow Cost Profile

The RAFT comparison isolates the motion source: an identical 13-feature Tier 0+Tier 1 pipeline, with codec MVs swapped for dense RAFT-Large flow at 256×256 , average-pooled onto the 32×32 MV-macroblock lattice. The headline gap is +2.3 AUC OOD on the matched subset (0.855 vs. 0.832) at ID parity. The cost profile behind that one-line summary is as follows. Pretrained RAFT-Large (C+T+SKHT+V2 preset, 5.26M parameters) at 256×256 on the 11 consecutive pairs per clip costs ~ 400 ms per clip on an A40 GPU, about 20 GPU-min per 3,000-clip generator, whereas codec MVs are decoded for free from the bitstream. The gap is consistent with codec MVs being a coarser, sparser, energy-minimized proxy for true motion: replacing them with dense pretrained flow buys OOD robustness on top of the same spectral readout but does not buy ID accuracy.